QuantiFi 2026 - Exploratory Data Analysis Report

Training days -2, -1, 0 · 26 instruments · prepared 24 Oct 2026 following *quantifi_2026_eda_plan.md* (modules M1-M4; M0 partially, see Open questions).

Executive summary

- **Days are continuous.** Each day opens where the previous one closed (e.g. BLACK_PEARLS 11,880 -> 11,876; 12,111.5 -> 12,112), so day 1 should start near day 0's close. Carry state, not daily resets.
- **Raw mid is the best real-time fair value** for every product except AMETHYST, GOLDEN_PEARLS (flat regimes), PAPER_LANTERNS and TOFFEE_CAKE (short EWMA). Weighted mid and wall mid are worse at every horizon; book volumes carry no information.
- **Order-book imbalance has no predictive power.** |corr(L1 or L3 imbalance, next-tick change)| < 0.035 everywhere, and P(next move up | bid-heavy) is about 0.50.
- **Market trades are uninformed.** 100% of prints sit at or through the same-tick touch (50% at ask, 46% at bid, 4% through), and aggressors lose ~half the spread at 1, 5 and 20 ticks on almost every product (exception: TELLURIUM, whose flow is one-sided on trending days).
- **The edge is structural and in spread capture**: AMETHYST/GOLDEN/PAPER_LANTERNS/TOFFEE making, PINK-vs-pearl-sum taking, and voucher relative value. TELLURIUM is the main risk to control.

Product classification

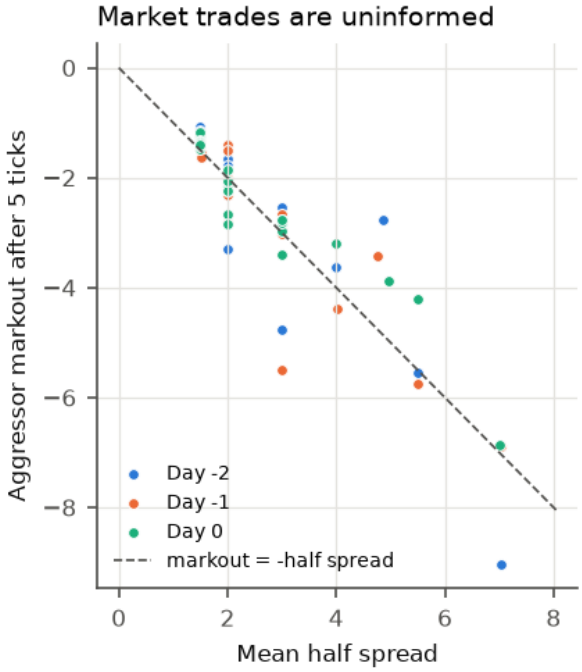
Product	Limit	Archetype	Fair value	Candidate strategy
AMETHYST	50	Fixed fair value (10,000)	Constant 10,000 (or EWMA span >= 200)	Market make around 10,000; take any ask < 10,000 / bid > 10,000
GOLDEN_PEARLS	20	Piecewise-constant fair value with step-downs	Current plateau (long EWMA), reset on a confirmed step	AMETHYST-style making around the plateau; cut inventory on step signals
PINK_PEARLS	20	Plateaus + jumps; the 'plug' that restores the pearl total	Plateau mid, adjusted toward 50,000 - sum(other pearl mids)	Take PINK in the restoring direction when |pearl sum - 50,000| > 10
PAPER_LANTERNS	50	Slow-drifting fair value plus i.i.d. quote noise	EWMA span ~5 of mid (~16% lower RMSE than raw mid)	Market make around the smoothed mid; fade 1-tick mid jumps
TOFFEE_CAKE	50	Mean-reverting around ~5,000, very wide spread	Short EWMA (span ~3) pulled toward a long-run anchor near 5,000	Passive making inside the 14-wide spread; lean against deviations from 5,000
BLUE_PEARLS	20	Trending with short-horizon momentum	Mid plus a momentum adjustment	Skew quotes with the 10-tick trend; avoid fading
WHITE_PEARLS	20	Random walk; pearl-sum member	Raw mid	Standalone making is marginal; part of the pearl-sum constraint
BLACK_PEARLS	20	Random walk; pearl-sum member	Raw mid	Passive making at/inside the touch (spread 8)
JELLY	50	Random walk, wide spread (~11)	Raw mid	Passive making for spread capture with tight inventory control
PEANUT_BUTTER	50	High-volatility random walk	Raw mid	Small passive making, aggressive inventory skew
TELLURIUM	40	Jump-diffusion; largest and most frequent risk events	Raw mid; reset immediately after jumps	Avoid resting size; trade small, wide, and only with jump protection
LYCHEE	200	Random walk; underlying for the vouchers	Raw mid	Make modestly; primary use is delta hedging the voucher book (limit 200)
DRAGONFRUIT	40	Random walk; basket leg (weight ~2)	Raw mid	Hedge leg for basket exposure; limited standalone edge
GRANADILLA	40	Random walk; basket leg (weight ~3)	Raw mid	Hedge leg; passive making at touch is mildly positive
MANGOSTEEN	40	Random walk; basket leg (weight ~1)	Raw mid	Hedge leg; occasional passive making
FRUIT_BASKET	40	Basket = 2*DRAGONFRUIT + 3*GRANADILLA + 1*MANGOSTEEN + drifting premium	Synthetic basket plus slowly-updated premium	Use component prices as a fair-value input; stat-arb on the premium is not supported
VOUCHER_6800..8100	50	Calls on LYCHEE; smile, IV > RV	BS with fitted smile	Short vol / butterfly RV, delta-hedged with LYCHEE

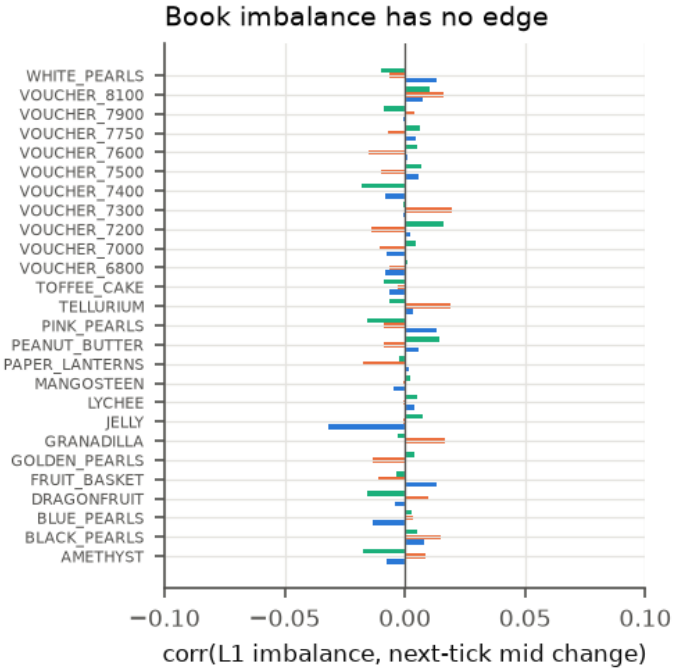
Data audit

- 3 days x 26 products x 10,000 ticks = 780,000 snapshots; no duplicate (day, timestamp, product); no missing L1 bid or ask; no crossed books.
- L2 is missing ~9% of the time, uniformly across products; normalise depth features.
- Ask volumes in the CSVs are POSITIVE (the wiki's OrderDepth example uses negative ask volumes); normalise with abs() in trader.py.
- Trades CSVs have no day column; the day is taken from the file name. buyer/seller are always blank.
- Voucher symbols are VOUCHER_<strike> (not LYCHEE_VOUCHERS_*); the mangosteen symbol is MANGOSTEEN.
- profit_and_loss is always 0 and was ignored.

Method

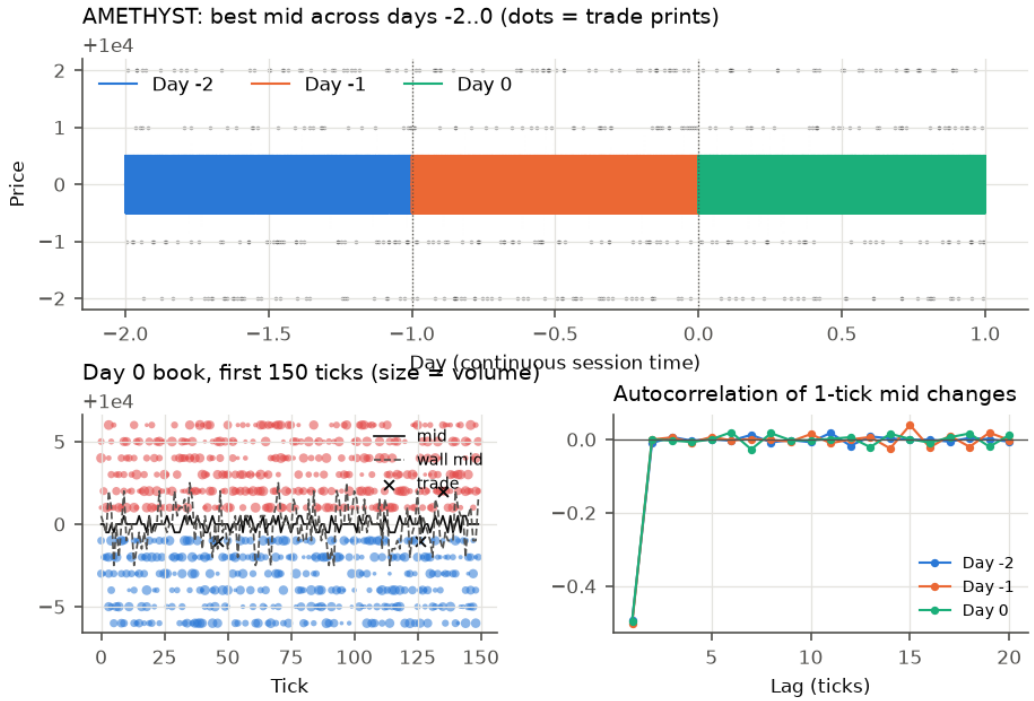
For each product and day: price path and spread statistics; autocorrelation and variance ratios of mid changes; RMSE of real-time fair-value estimators (mid, volume-weighted mid, wall mid, EWMA spans 3-200) against the mid 1/10/50 ticks ahead; conditional forward returns on EWMA deviation, 10-tick momentum and L1/L3 imbalance; trade classification against the same-timestamp book and aggressor markouts; and a 1-lot passive-quote replay under the wiki's fill rule at four offsets from the touch (EV = sum of 20-tick markouts per day, no inventory limits). Structural families add basket regressions, pearl-sum stationarity, and a Black-Scholes IV surface with no-arbitrage checks.





AMETHYST (position limit 50)

Archetype: Fixed fair value (10,000). **Fair value:** Constant 10,000 (or EWMA span >= 200). **Candidate strategy:** Market make around 10,000; take any ask < 10,000 / bid > 10,000.



Key findings

- Mid never leaves 9,999.5-10,000.5 on any day; 1-tick mid changes have ACF(1) = -0.50, the signature of pure noise around a constant.
- A long EWMA cuts 10-tick fair-value RMSE by ~29% vs raw mid; wall mid and weighted mid are worse.
- Spread is 3 on average. Passive replay: quoting AT the touch earns ~1.3/lot over ~100-120 fills/day (best EV); improving by 2 ticks turns negative.

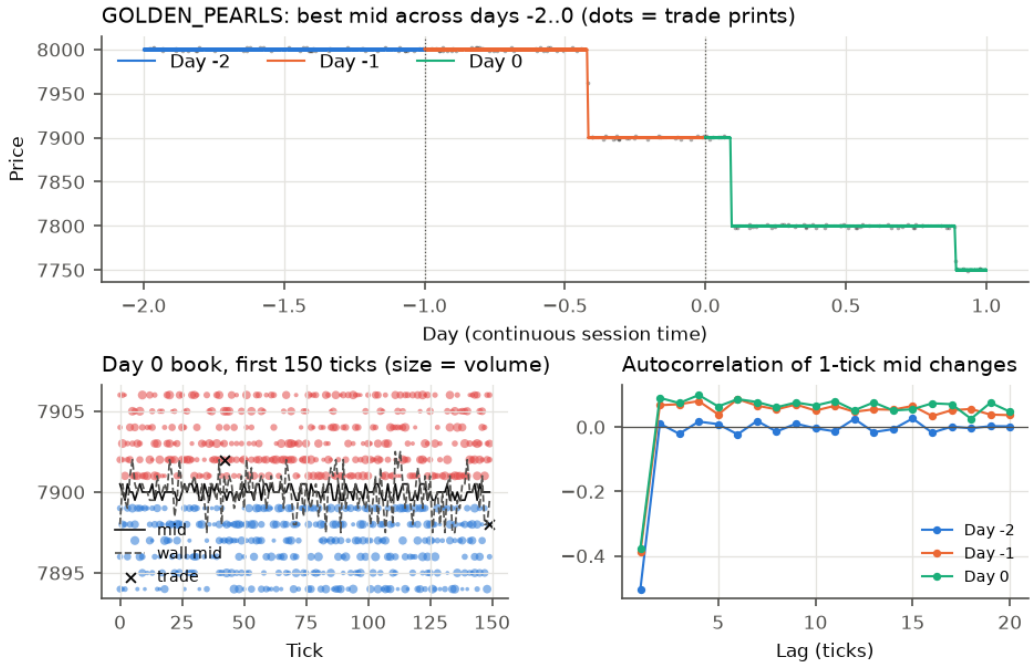
Findings by day

- **Day -2:** fell 0.5 (10,000.0 -> 9,999.5, range 9,999.5-10,000.5); spread 3.0, tick sigma 0.49; reversal (corr -0.51); strong noise bounce (ACF -0.49). 136 prints / 358 lots; aggressor 5-tick markout -1.5. Best passive quote: at touch (+135 per lot-day).
- **Day -1:** was flat 0.0 (10,000.0 -> 10,000.0, range 9,999.5-10,000.5); spread 3.0, tick sigma 0.51; reversal (corr -0.49); strong noise bounce (ACF -0.50). 159 prints / 396 lots; aggressor 5-tick markout -1.5. Best passive quote: at touch (+159 per lot-day).
- **Day 0:** fell 0.5 (10,000.0 -> 9,999.5, range 9,999.5-10,000.5); spread 3.0, tick sigma 0.49; reversal (corr -0.50); strong noise bounce (ACF -0.50). 135 prints / 323 lots; aggressor 5-tick markout -1.5. Best passive quote: at touch (+132 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	10,000.0 -> 9,999.5 (-0.5)	10,000.0 -> 10,000.0 (+0.0)	10,000.0 -> 9,999.5 (-0.5)
Low - high	9,999.5 - 10,000.5	9,999.5 - 10,000.5	9,999.5 - 10,000.5
Mean spread / L1 depth	3.00 / 17.5	3.01 / 17.4	2.99 / 17.4
Tick sigma / zero-move %	0.49 / 38%	0.51 / 37%	0.49 / 38%
ACF(1) mid / wall mid	-0.49 / -0.50	-0.50 / -0.50	-0.50 / -0.51
Jumps > 5 sigma (max move)	0 (1.0)	0 (1.0)	0 (1.0)
Corr momentum10 / EWMA dev -> fwd10	-0.51 / -0.71	-0.49 / -0.70	-0.50 / -0.71
Corr L1 imbalance -> next tick	-0.008	+0.009	-0.017
10-tick fair-value RMSE	mid 0.50; best span 200 -> 0.35	mid 0.50; best span 200 -> 0.36	mid 0.50; best span 200 -> 0.35
Trades / lots (avg size)	136 / 358 (2.6)	159 / 396 (2.5)	135 / 323 (2.4)
Prints at ask / bid	44% / 56%	48% / 52%	51% / 49%
Aggressor markout h1/h5/h20	-1.5 / -1.5 / -1.4	-1.5 / -1.5 / -1.5	-1.5 / -1.5 / -1.4
Best 1-lot passive quote (replay)	at touch: 105 fills, +1.3/lot, EV +135	at touch: 120 fills, +1.3/lot, EV +159	at touch: 101 fills, +1.3/lot, EV +132

GOLDEN_PEARLS (position limit 20)

Archetype: Piecewise-constant fair value with step-downs. **Fair value:** Current plateau (long EWMA), reset on a confirmed step. **Candidate strategy:** AMETHYST-style making around the plateau; cut inventory on step signals.



Key findings

- Day -2: flat at 8,000 exactly like AMETHYST (ACF(1) -0.50). Day -1: steps down to 7,900 mid-session. Day 0: 7,900 -> 7,800 early and drifts to ~7,750 by the close.
- All 35 moves > 2 over days -1/0 were DOWN, and the price kept falling ~33 over the next 20 ticks, so a passive bid during a step is adversely selected.
- Quoting at the touch still earns ~1.1-1.4/lot on 34-50 fills/day.

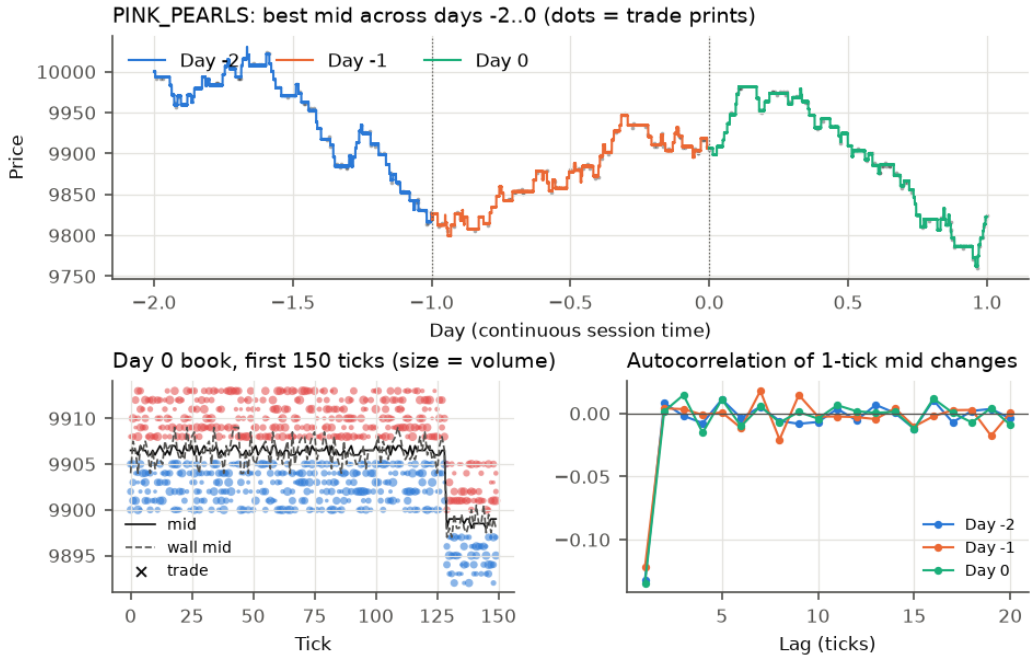
Findings by day

- **Day -2:** rose 0.5 (7,999.5 -> 8,000.0, range 7,999.5-8,000.5); spread 3.0, tick sigma 0.50; reversal (corr -0.50); strong noise bounce (ACF -0.50). 49 prints / 98 lots; aggressor 5-tick markout -1.4. Best passive quote: at touch (+48 per lot-day).
- **Day -1:** fell 101.0 (8,000.5 -> 7,899.5, range 7,899.5-8,000.5); spread 3.0, tick sigma 0.52; momentum (corr +0.70); strong noise bounce (ACF -0.39). 62 prints / 121 lots; aggressor 5-tick markout -1.6. Best passive quote: improve 1 (+76 per lot-day).
- **Day 0:** fell 151.0 (7,900.5 -> 7,749.5, range 7,749.5-7,900.5); spread 3.0, tick sigma 0.53; momentum (corr +0.72); strong noise bounce (ACF -0.38). 66 prints / 130 lots; aggressor 5-tick markout -1.4. Best passive quote: at touch (+56 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	7,999.5 -> 8,000.0 (+0.5)	8,000.5 -> 7,899.5 (-101.0)	7,900.5 -> 7,749.5 (-151.0)
Low - high	7,999.5 - 8,000.5	7,899.5 - 8,000.5	7,749.5 - 7,900.5
Mean spread / L1 depth	3.00 / 15.0	3.02 / 15.0	2.99 / 15.0
Tick sigma / zero-move %	0.50 / 38%	0.52 / 38%	0.53 / 37%
ACF(1) mid / wall mid	-0.50 / -0.49	-0.39 / -0.50	-0.38 / -0.49
Jumps > 5 sigma (max move)	0 (1.0)	2 (3.0)	4 (3.0)
Corr momentum10 / EWMA dev -> fwd10	-0.50 / -0.70	+0.70 / +0.47	+0.72 / +0.47
Corr L1 imbalance -> next tick	+0.000	-0.014	+0.004
10-tick fair-value RMSE	mid 0.50; best span 200 -> 0.36	mid 1.46; best span 1 -> 1.46	mid 1.61; best span 1 -> 1.61
Trades / lots (avg size)	49 / 98 (2.0)	62 / 121 (2.0)	66 / 130 (2.0)
Prints at ask / bid	51% / 49%	52% / 48%	59% / 41%
Aggressor markout h1/h5/h20	-1.4 / -1.4 / -1.5	-1.6 / -1.6 / -2.2	-1.6 / -1.4 / -1.1
Best 1-lot passive quote (replay)	at touch: 34 fills, +1.4/lot, EV +48	improve 1: 62 fills, +1.2/lot, EV +76	at touch: 50 fills, +1.1/lot, EV +56

PINK_PEARLS (position limit 20)

Archetype: Plateaus + jumps; the 'plug' that restores the pearl total. **Fair value:** Plateau mid, adjusted toward 50,000 - sum(other pearl mids). **Candidate strategy:** Take PINK in the restoring direction when |pearl sum - 50,000| > 10.



Key findings

- ~62 jumps/day of ~10.7 with flat noise (sigma 0.5) in between. 100% of jumps on every day move the five-pearl total back toward 50,000.
- Jump size is almost exactly minus the deviation (slope -1.10 to -1.12, residual sd ~1.3): a jump resets the total to 50,000. Jump probability per tick is 5-11% when |sum - 50,000| > 10 vs 0.4% otherwise.
- Mean 20-tick PINK move: +4.7 to +6.8 when the sum is < -10, -6.6 to -10.2 when > +10: several times the ~2 half spread.

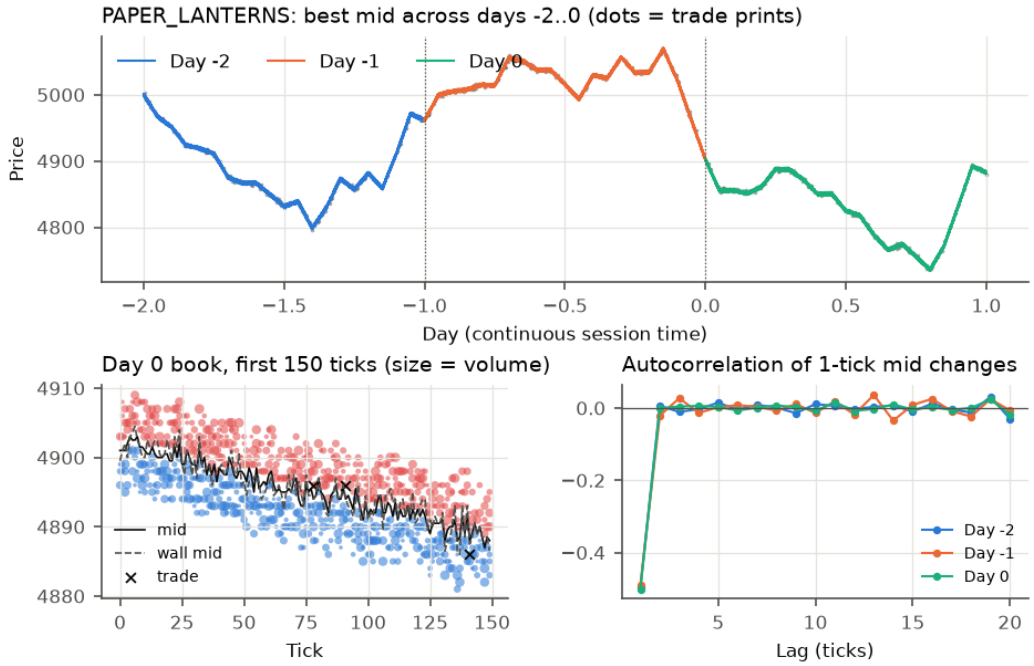
Findings by day

- **Day -2:** fell 173.5 (10,000.0 -> 9,826.5, range 9,816.0-10,030.0); spread 4.0, tick sigma 1.00; 62 jumps > 5 sigma. 61 prints / 126 lots; aggressor 5-tick markout -1.7. Best passive quote: improve 1 (+62 per lot-day).
- **Day -1:** rose 80.0 (9,826.0 -> 9,906.0, range 9,799.0-9,947.0); spread 4.0, tick sigma 1.00; 63 jumps > 5 sigma. 61 prints / 130 lots; aggressor 5-tick markout -2.2. Best passive quote: improve 1 (+92 per lot-day).
- **Day 0:** fell 83.5 (9,906.5 -> 9,823.0, range 9,762.0-9,982.0); spread 4.0, tick sigma 1.00; 63 jumps > 5 sigma. 62 prints / 134 lots; aggressor 5-tick markout -2.2. Best passive quote: at touch (+95 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	10,000.0 -> 9,826.5 (-173.5)	9,826.0 -> 9,906.0 (+80.0)	9,906.5 -> 9,823.0 (-83.5)
Low - high	9,816.0 - 10,030.0	9,799.0 - 9,947.0	9,762.0 - 9,982.0
Mean spread / L1 depth	3.99 / 15.0	3.99 / 15.0	4.00 / 15.0
Tick sigma / zero-move %	1.00 / 37%	1.00 / 37%	1.00 / 38%
ACF(1) mid / wall mid	-0.13 / -0.41	-0.12 / -0.39	-0.13 / -0.41
Jumps > 5 sigma (max move)	62 (15.0)	63 (15.5)	63 (15.5)
Corr momentum10 / EWMA dev -> fwd10	-0.03 / -0.01	-0.03 / -0.03	-0.02 / +0.02
Corr L1 imbalance -> next tick	+0.013	-0.009	-0.016
10-tick fair-value RMSE	mid 2.78; best span 1 -> 2.78	mid 2.82; best span 1 -> 2.82	mid 2.76; best span 1 -> 2.76
Trades / lots (avg size)	61 / 126 (2.1)	61 / 130 (2.1)	62 / 134 (2.2)
Prints at ask / bid	56% / 44%	39% / 61%	47% / 53%
Aggressor markout h1/h5/h20	-1.9 / -1.7 / -1.9	-2.1 / -2.2 / -2.5	-1.9 / -2.2 / -2.2
Best 1-lot passive quote (replay)	improve 1: 61 fills, +1.0/lot, EV +62	improve 1: 61 fills, +1.5/lot, EV +92	at touch: 46 fills, +2.1/lot, EV +95

PAPER_LANTERNS (position limit 50)

Archetype: Slow-drifting fair value plus i.i.d. quote noise. **Fair value:** EWMA span ~5 of mid (~16% lower RMSE than raw mid). **Candidate strategy:** Market make around the smoothed mid; fade 1-tick mid jumps.



Key findings

- ACF(1) of mid changes = -0.50 on all three days and variance ratio at 10 ticks ~0.11: most tick-to-tick movement is noise that reverses immediately.
- The underlying level drifts down slowly (-37.5, -58, -19.5 per day). 10-tick momentum has corr -0.30 with the next 10-tick move.
- Quoting at the touch earns ~1.3-1.6/lot on 66-76 fills/day; improving by 2 ticks loses.

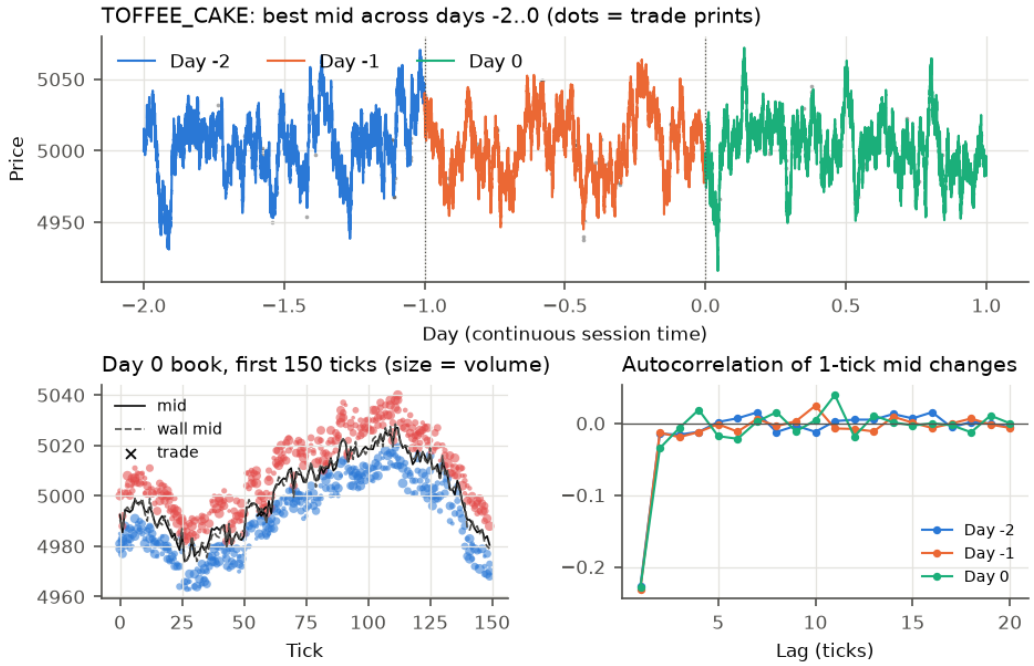
Findings by day

- **Day -2:** fell 37.5 (4,999.5 -> 4,962.0, range 4,797.0-5,001.5); spread 4.0, tick sigma 1.51; reversal (corr -0.30); strong noise bounce (ACF -0.50). 104 prints / 264 lots; aggressor 5-tick markout -1.9. Best passive quote: at touch (+100 per lot-day).
- **Day -1:** fell 58.0 (4,960.5 -> 4,902.5, range 4,901.0-5,071.0); spread 4.0, tick sigma 1.50; reversal (corr -0.30); strong noise bounce (ACF -0.49). 101 prints / 255 lots; aggressor 5-tick markout -2.0. Best passive quote: at touch (+118 per lot-day).
- **Day 0:** fell 19.5 (4,901.0 -> 4,881.5, range 4,734.0-4,903.0); spread 4.0, tick sigma 1.52; reversal (corr -0.32); strong noise bounce (ACF -0.50). 103 prints / 252 lots; aggressor 5-tick markout -1.8. Best passive quote: at touch (+93 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	4,999.5 -> 4,962.0 (-37.5)	4,960.5 -> 4,902.5 (-58.0)	4,901.0 -> 4,881.5 (-19.5)
Low - high	4,797.0 - 5,001.5	4,901.0 - 5,071.0	4,734.0 - 4,903.0
Mean spread / L1 depth	4.00 / 17.5	4.00 / 17.5	4.00 / 17.5
Tick sigma / zero-move %	1.51 / 13%	1.50 / 13%	1.52 / 13%
ACF(1) mid / wall mid	-0.50 / -0.49	-0.49 / -0.51	-0.50 / -0.50
Jumps > 5 sigma (max move)	0 (5.0)	0 (5.0)	0 (5.0)
Corr momentum10 / EWMA dev -> fwd10	-0.30 / +0.08	-0.30 / +0.10	-0.32 / +0.06
Corr L1 imbalance -> next tick	+0.002	-0.017	-0.002
10-tick fair-value RMSE	mid 1.61; best span 5 -> 1.35	mid 1.62; best span 5 -> 1.36	mid 1.61; best span 5 -> 1.34
Trades / lots (avg size)	104 / 264 (2.5)	101 / 255 (2.5)	103 / 252 (2.4)
Prints at ask / bid	41% / 59%	50% / 50%	52% / 48%
Aggressor markout h1/h5/h20	-2.0 / -1.9 / -2.0	-2.0 / -2.0 / -2.1	-1.9 / -1.8 / -2.1
Best 1-lot passive quote (replay)	at touch: 76 fills, +1.3/lot, EV +100	at touch: 72 fills, +1.6/lot, EV +118	at touch: 66 fills, +1.4/lot, EV +93

TOFFEE_CAKE (position limit 50)

Archetype: Mean-reverting around ~5,000, very wide spread. **Fair value:** Short EWMA (span ~3) pulled toward a long-run anchor near 5,000. **Candidate strategy:** Passive making inside the 14-wide spread; lean against deviations from 5,000.



Key findings

- Range 4,916-5,072 over three days; variance ratio at 1,000 ticks is only ~0.11, i.e. strong long-horizon mean reversion. ACF(1) of mid changes ~ -0.23.
- Widest spread in the universe (~14). Aggressors lose ~7 (the half spread): flow is uninformed.
- Passive replay is positive at every offset on every day; improving by 1-2 ticks gives the best EV (+88 to +309 per lot-day).

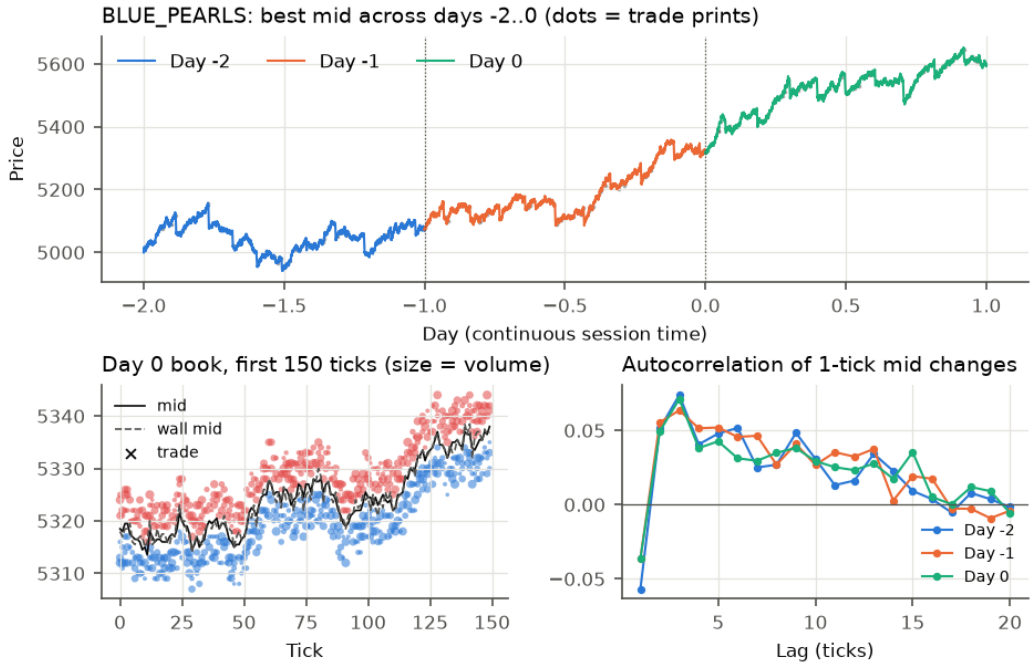
Findings by day

- **Day -2:** rose 36.0 (5,000.0 -> 5,036.0, range 4,931.0-5,070.5); spread 14.1, tick sigma 3.05; strong noise bounce (ACF -0.23). 41 prints / 157 lots; aggressor 5-tick markout -9.0. Best passive quote: improve 1 (+308 per lot-day).
- **Day -1:** fell 46.0 (5,036.5 -> 4,990.5, range 4,945.0-5,063.5); spread 14.1, tick sigma 3.11; strong noise bounce (ACF -0.23). 48 prints / 167 lots; aggressor 5-tick markout -6.9. Best passive quote: improve 2 (+184 per lot-day).
- **Day 0:** fell 7.5 (4,992.5 -> 4,985.0, range 4,916.0-5,072.0); spread 14.0, tick sigma 3.15; strong noise bounce (ACF -0.23). 46 prints / 148 lots; aggressor 5-tick markout -6.9. Best passive quote: improve 2 (+102 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	5,000.0 -> 5,036.0 (+36.0)	5,036.5 -> 4,990.5 (-46.0)	4,992.5 -> 4,985.0 (-7.5)
Low - high	4,931.0 - 5,070.5	4,945.0 - 5,063.5	4,916.0 - 5,072.0
Mean spread / L1 depth	14.09 / 17.5	14.06 / 17.6	14.00 / 17.6
Tick sigma / zero-move %	3.05 / 8%	3.11 / 8%	3.15 / 7%
ACF(1) mid / wall mid	-0.23 / -0.34	-0.23 / -0.33	-0.23 / -0.33
Jumps > 5 sigma (max move)	1 (15.5)	0 (14.0)	0 (14.0)
Corr momentum10 / EWMA dev -> fwd10	-0.03 / -0.06	-0.06 / -0.09	-0.02 / -0.04
Corr L1 imbalance -> next tick	-0.007	-0.003	-0.009
10-tick fair-value RMSE	mid 7.10; best span 3 -> 7.08	mid 7.05; best span 3 -> 7.03	mid 7.17; best span 3 -> 7.15
Trades / lots (avg size)	41 / 157 (3.8)	48 / 167 (3.5)	46 / 148 (3.2)
Prints at ask / bid	41% / 59%	56% / 44%	59% / 41%
Aggressor markout h1/h5/h20	-7.4 / -9.0 / -11.4	-7.0 / -6.9 / -7.5	-6.9 / -6.9 / -5.5
Best 1-lot passive quote (replay)	improve 1: 30 fills, +10.6/lot, EV +308	improve 2: 41 fills, +4.5/lot, EV +184	improve 2: 29 fills, +3.5/lot, EV +102

BLUE_PEARLS (position limit 20)

Archetype: Trending with short-horizon momentum. **Fair value:** Mid plus a momentum adjustment. **Candidate strategy:** Skew quotes with the 10-tick trend; avoid fading.



Key findings

- Up every day: +72, +247, +277 (5,000 -> 5,595).
- Variance ratio 1.26-1.31 at 10 ticks and 1.5-1.8 at 100 ticks; 10-tick momentum has corr +0.22 to +0.24 with the next 10-tick move, the only consistent momentum signal found.
- Passive at-touch markout ~+1.2-1.3/lot; spread 6.

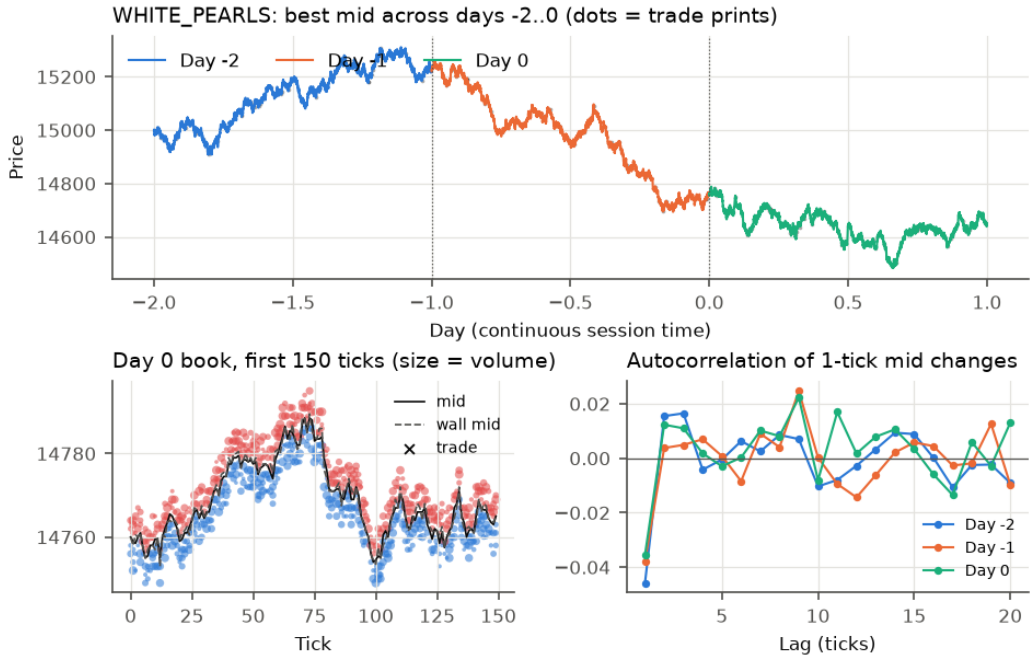
Findings by day

- **Day -2:** rose 71.5 (5,000.0 -> 5,071.5, range 4,941.0-5,157.5); spread 6.0, tick sigma 1.42; momentum (corr +0.22). 50 prints / 94 lots; aggressor 5-tick markout -2.8. Best passive quote: improve 1 (+52 per lot-day).
- **Day -1:** rose 246.5 (5,072.0 -> 5,318.5, range 5,070.5-5,359.0); spread 6.0, tick sigma 1.41; momentum (corr +0.24). 63 prints / 130 lots; aggressor 5-tick markout -2.8. Best passive quote: 1 behind (+58 per lot-day).
- **Day 0:** rose 276.5 (5,318.5 -> 5,595.0, range 5,313.5-5,651.5); spread 6.0, tick sigma 1.41; momentum (corr +0.23). 59 prints / 124 lots; aggressor 5-tick markout -2.8. Best passive quote: improve 1 (+66 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	5,000.0 -> 5,071.5 (+71.5)	5,072.0 -> 5,318.5 (+246.5)	5,318.5 -> 5,595.0 (+276.5)
Low - high	4,941.0 - 5,157.5	5,070.5 - 5,359.0	5,313.5 - 5,651.5
Mean spread / L1 depth	5.99 / 15.0	6.00 / 15.0	6.00 / 15.0
Tick sigma / zero-move %	1.42 / 14%	1.41 / 14%	1.41 / 14%
ACF(1) mid / wall mid	-0.06 / -0.33	-0.04 / -0.31	-0.04 / -0.30
Jumps > 5 sigma (max move)	1 (7.5)	0 (6.5)	0 (6.5)
Corr momentum10 / EWMA dev -> fwd10	+0.22 / +0.10	+0.24 / +0.06	+0.23 / +0.11
Corr L1 imbalance -> next tick	-0.014	+0.003	+0.003
10-tick fair-value RMSE	mid 5.02; best span 1 -> 5.02	mid 5.11; best span 1 -> 5.11	mid 5.03; best span 1 -> 5.03
Trades / lots (avg size)	50 / 94 (1.9)	63 / 130 (2.1)	59 / 124 (2.1)
Prints at ask / bid	46% / 54%	51% / 49%	49% / 51%
Aggressor markout h1/h5/h20	-2.9 / -2.8 / -2.7	-3.0 / -2.8 / -2.0	-2.8 / -2.8 / -2.3
Best 1-lot passive quote (replay)	improve 1: 44 fills, +1.2/lot, EV +52	1 behind: 27 fills, +2.1/lot, EV +58	improve 1: 53 fills, +1.2/lot, EV +66

WHITE_PEARLS (position limit 20)

Archetype: Random walk; pearl-sum member. **Fair value:** Raw mid. **Candidate strategy:** Standalone making is marginal; part of the pearl-sum constraint.



Key findings

- Days +220, -460, -110. Strong negative return correlation with BLACK_PEARLS (-0.60).
- The pearl-sum deviation does not predict WHITE's own returns (|corr| < 0.04).
- Passive replay is roughly break-even; limit only 20.

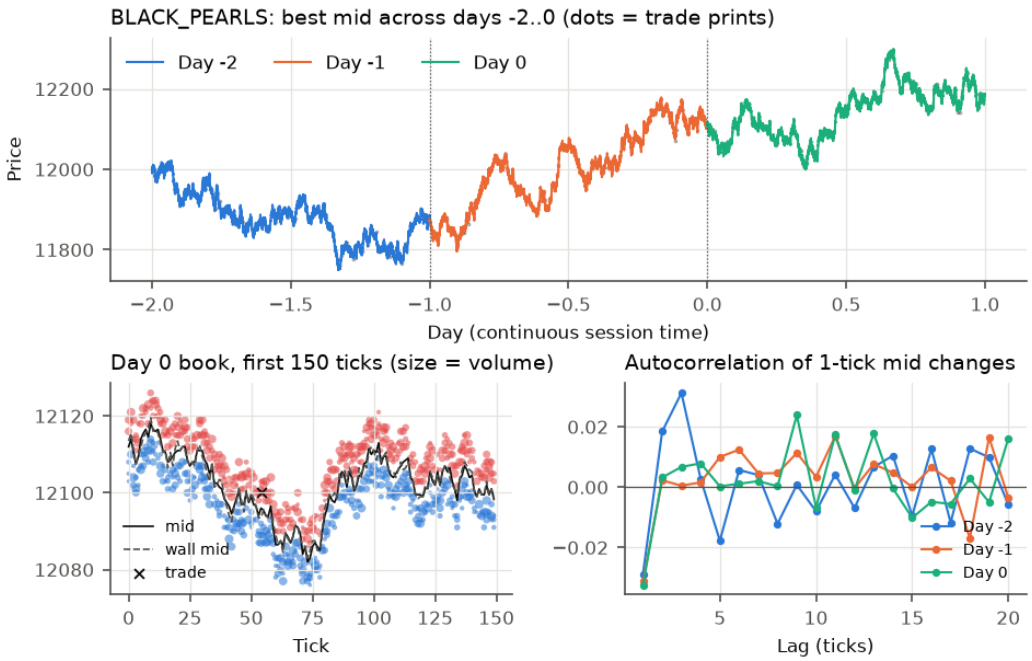
Findings by day

- **Day -2:** rose 219.5 (14,999.5 -> 15,219.0, range 14,908.5-15,311.0); spread 4.0, tick sigma 2.27; no material autocorrelation signal. 67 prints / 138 lots; aggressor 5-tick markout -1.8. Best passive quote: 1 behind (+26 per lot-day).
- **Day -1:** fell 460.0 (15,220.5 -> 14,760.5, range 14,694.0-15,259.0); spread 4.0, tick sigma 2.25; no material autocorrelation signal. 61 prints / 108 lots; aggressor 5-tick markout -1.5. Best passive quote: at touch (+28 per lot-day).
- **Day 0:** fell 110.5 (14,760.0 -> 14,649.5, range 14,483.5-14,788.5); spread 4.0, tick sigma 2.24; no material autocorrelation signal. 66 prints / 133 lots; aggressor 5-tick markout -2.1. Best passive quote: at touch (+30 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	14,999.5 -> 15,219.0 (+219.5)	15,220.5 -> 14,760.5 (-460.0)	14,760.0 -> 14,649.5 (-110.5)
Low - high	14,908.5 - 15,311.0	14,694.0 - 15,259.0	14,483.5 - 14,788.5
Mean spread / L1 depth	3.99 / 15.0	4.01 / 15.0	4.01 / 15.0
Tick sigma / zero-move %	2.27 / 9%	2.25 / 9%	2.24 / 9%
ACF(1) mid / wall mid	-0.05 / -0.20	-0.04 / -0.21	-0.04 / -0.19
Jumps > 5 sigma (max move)	0 (8.5)	0 (10.0)	0 (8.5)
Corr momentum10 / EWMA dev -> fwd10	+0.01 / -0.01	+0.01 / +0.01	+0.05 / +0.05
Corr L1 imbalance -> next tick	+0.013	-0.006	-0.010
10-tick fair-value RMSE	mid 7.07; best span 1 -> 7.07	mid 7.00; best span 1 -> 7.00	mid 7.04; best span 1 -> 7.04
Trades / lots (avg size)	67 / 138 (2.1)	61 / 108 (1.8)	66 / 133 (2.0)
Prints at ask / bid	46% / 54%	62% / 38%	56% / 44%
Aggressor markout h1/h5/h20	-1.6 / -1.8 / -2.0	-2.0 / -1.5 / -2.9	-1.8 / -2.1 / -2.8
Best 1-lot passive quote (replay)	1 behind: 30 fills, +0.9/lot, EV +26	at touch: 50 fills, +0.6/lot, EV +28	at touch: 42 fills, +0.7/lot, EV +30

BLACK_PEARLS (position limit 20)

Archetype: Random walk; pearl-sum member. **Fair value:** Raw mid. **Candidate strategy:** Passive making at/inside the touch (spread 8).



Key findings

- Days -120, +236, +72. Return corr -0.60 with WHITE and -0.40 with BLUE.
- Spread ~8 gives the best per-fill markout among pearls (+2 to +4.5 per lot when improving by 1).
- The 20-lot limit caps total opportunity.

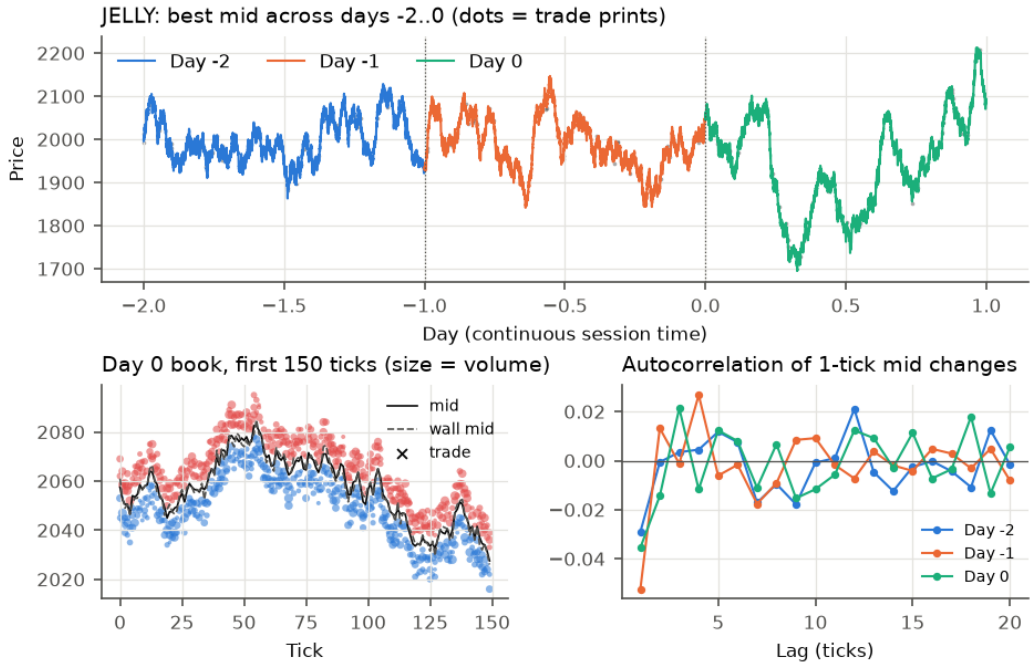
Findings by day

- **Day -2:** fell 119.5 (11,999.5 -> 11,880.0, range 11,749.5-12,022.0); spread 8.0, tick sigma 2.23; no material autocorrelation signal. 55 prints / 105 lots; aggressor 5-tick markout -3.6. Best passive quote: improve 2 (+194 per lot-day).
- **Day -1:** rose 235.5 (11,876.0 -> 12,111.5, range 11,795.5-12,179.0); spread 8.0, tick sigma 2.26; no material autocorrelation signal. 51 prints / 91 lots; aggressor 5-tick markout -4.4. Best passive quote: improve 1 (+81 per lot-day).
- **Day 0:** rose 72.0 (12,112.0 -> 12,184.0, range 11,999.5-12,299.0); spread 8.0, tick sigma 2.22; no material autocorrelation signal. 58 prints / 112 lots; aggressor 5-tick markout -3.2. Best passive quote: improve 1 (+190 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	11,999.5 -> 11,880.0 (-119.5)	11,876.0 -> 12,111.5 (+235.5)	12,112.0 -> 12,184.0 (+72.0)
Low - high	11,749.5 - 12,022.0	11,795.5 - 12,179.0	11,999.5 - 12,299.0
Mean spread / L1 depth	8.00 / 15.0	8.01 / 15.0	8.01 / 15.0
Tick sigma / zero-move %	2.23 / 9%	2.26 / 9%	2.22 / 9%
ACF(1) mid / wall mid	-0.03 / -0.20	-0.03 / -0.20	-0.03 / -0.20
Jumps > 5 sigma (max move)	0 (8.0)	0 (8.0)	0 (9.5)
Corr momentum10 / EWMA dev -> fwd10	+0.00 / -0.02	+0.06 / +0.03	+0.04 / +0.04
Corr L1 imbalance -> next tick	+0.008	+0.015	+0.005
10-tick fair-value RMSE	mid 7.09; best span 1 -> 7.09	mid 7.07; best span 1 -> 7.07	mid 6.94; best span 1 -> 6.94
Trades / lots (avg size)	55 / 105 (1.9)	51 / 91 (1.8)	58 / 112 (1.9)
Prints at ask / bid	38% / 62%	61% / 39%	45% / 55%
Aggressor markout h1/h5/h20	-3.6 / -3.6 / -6.6	-4.1 / -4.4 / -4.6	-4.1 / -3.2 / -5.7
Best 1-lot passive quote (replay)	improve 2: 47 fills, +4.1/lot, EV +194	improve 1: 40 fills, +2.0/lot, EV +81	improve 1: 47 fills, +4.0/lot, EV +190

JELLY (position limit 50)

Archetype: Random walk, wide spread (~11). **Fair value:** Raw mid. **Candidate strategy:** Passive making for spread capture with tight inventory control.



Key findings

- Mid changes are close to uncorrelated (ACF(1) -0.03 to -0.05); variance ratios ~1 up to 100 ticks.
- Half spread ~5.5; aggressor markout ~-5.2 to -6.0 at 1 tick, so captured spread is not given back quickly.
- 1-lot replay EV swings with the daily trend (+146 on day -1, -121 on day 0): inventory risk, not adverse selection, dominates - use a reservation price r = mid - lambda * position.

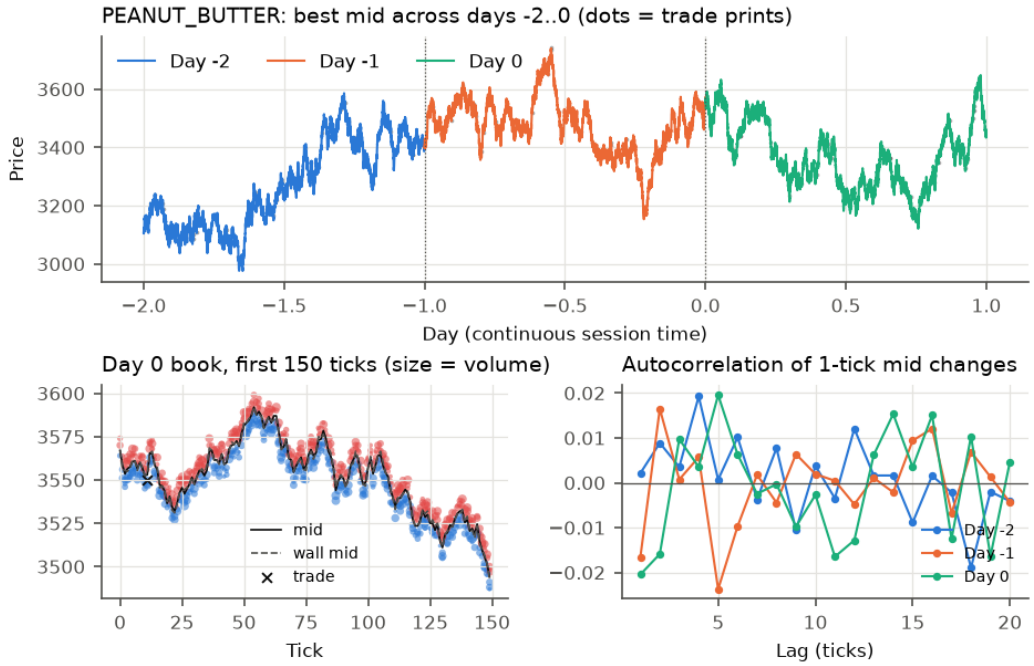
Findings by day

- **Day -2:** fell 67.0 (2,000.5 -> 1,933.5, range 1,863.0-2,128.5); spread 11.0, tick sigma 3.18; no material autocorrelation signal. 59 prints / 159 lots; aggressor 5-tick markout -5.6. Best passive quote: improve 2 (+109 per lot-day).
- **Day -1:** rose 121.5 (1,933.5 -> 2,055.0, range 1,841.5-2,147.0); spread 11.0, tick sigma 3.12; no material autocorrelation signal. 68 prints / 171 lots; aggressor 5-tick markout -5.7. Best passive quote: improve 2 (+218 per lot-day).
- **Day 0:** rose 32.5 (2,057.5 -> 2,090.0, range 1,695.0-2,212.5); spread 11.0, tick sigma 3.10; no material autocorrelation signal. 63 prints / 156 lots; aggressor 5-tick markout -4.2. Best passive quote: improve 1 (-106 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	2,000.5 -> 1,933.5 (-67.0)	1,933.5 -> 2,055.0 (+121.5)	2,057.5 -> 2,090.0 (+32.5)
Low - high	1,863.0 - 2,128.5	1,841.5 - 2,147.0	1,695.0 - 2,212.5
Mean spread / L1 depth	11.02 / 15.0	11.00 / 15.1	11.02 / 15.0
Tick sigma / zero-move %	3.18 / 6%	3.12 / 7%	3.10 / 6%
ACF(1) mid / wall mid	-0.03 / -0.20	-0.05 / -0.22	-0.04 / -0.22
Jumps > 5 sigma (max move)	0 (11.5)	0 (12.0)	0 (11.5)
Corr momentum10 / EWMA dev -> fwd10	-0.02 / -0.01	-0.00 / +0.05	-0.01 / +0.08
Corr L1 imbalance -> next tick	-0.032	-0.000	+0.007
10-tick fair-value RMSE	mid 9.84; best span 1 -> 9.84	mid 9.55; best span 1 -> 9.55	mid 9.52; best span 1 -> 9.52
Trades / lots (avg size)	59 / 159 (2.7)	68 / 171 (2.5)	63 / 156 (2.5)
Prints at ask / bid	54% / 46%	54% / 46%	56% / 44%
Aggressor markout h1/h5/h20	-5.8 / -5.6 / -5.9	-5.7 / -5.7 / -7.4	-5.0 / -4.2 / +0.6
Best 1-lot passive quote (replay)	improve 2: 49 fills, +2.2/lot, EV +109	improve 2: 57 fills, +3.8/lot, EV +218	improve 1: 41 fills, -2.6/lot, EV -106

PEANUT_BUTTER (position limit 50)

Archetype: High-volatility random walk. **Fair value:** Raw mid. **Candidate strategy:** Small passive making, aggressive inventory skew.



Key findings

- Tick sigma ~4.8 against a spread of only ~6: volatility is high relative to the edge.
- No autocorrelation or imbalance signal on any day.
- Passive replay is negative on days -2 and -1 (-64 to -370) and positive on day 0; the sign follows the trend. Lowest-priority making candidate among the single products.

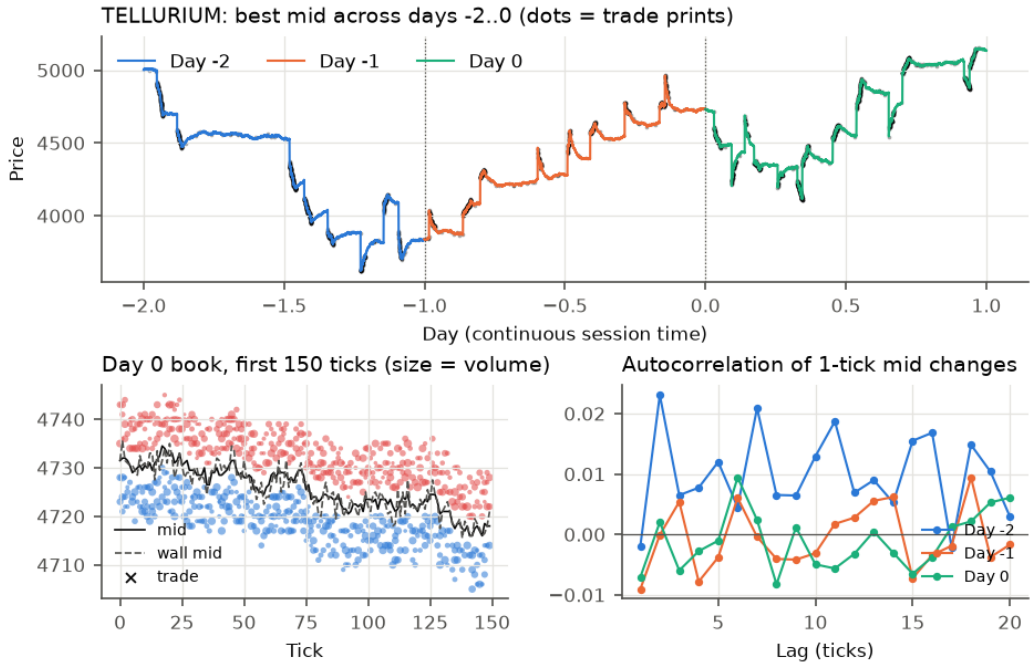
Findings by day

- **Day -2:** rose 297.5 (3,106.0 -> 3,403.5, range 2,976.0-3,586.0); spread 6.0, tick sigma 4.82; no material autocorrelation signal. 79 prints / 194 lots; aggressor 5-tick markout -2.5. Best passive quote: 1 behind (-86 per lot-day).
- **Day -1:** rose 164.0 (3,403.5 -> 3,567.5, range 3,153.5-3,739.5); spread 6.0, tick sigma 4.76; no material autocorrelation signal. 58 prints / 153 lots; aggressor 5-tick markout -2.7. Best passive quote: 1 behind (-64 per lot-day).
- **Day 0:** fell 129.0 (3,567.0 -> 3,438.0, range 3,121.0-3,648.0); spread 6.0, tick sigma 4.79; no material autocorrelation signal. 59 prints / 148 lots; aggressor 5-tick markout -3.0. Best passive quote: at touch (+230 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	3,106.0 -> 3,403.5 (+297.5)	3,403.5 -> 3,567.5 (+164.0)	3,567.0 -> 3,438.0 (-129.0)
Low - high	2,976.0 - 3,586.0	3,153.5 - 3,739.5	3,121.0 - 3,648.0
Mean spread / L1 depth	5.99 / 15.0	6.01 / 15.0	6.01 / 15.0
Tick sigma / zero-move %	4.82 / 4%	4.76 / 5%	4.79 / 4%
ACF(1) mid / wall mid	+0.00 / -0.04	-0.02 / -0.06	-0.02 / -0.07
Jumps > 5 sigma (max move)	0 (17.5)	0 (19.0)	0 (20.0)
Corr momentum10 / EWMA dev -> fwd10	+0.01 / -0.03	-0.00 / +0.03	-0.01 / +0.02
Corr L1 imbalance -> next tick	+0.006	-0.009	+0.014
10-tick fair-value RMSE	mid 15.65; best span 1 -> 15.65	mid 14.85; best span 1 -> 14.85	mid 14.97; best span 1 -> 14.97
Trades / lots (avg size)	79 / 194 (2.5)	58 / 153 (2.6)	59 / 148 (2.5)
Prints at ask / bid	53% / 47%	59% / 41%	59% / 41%
Aggressor markout h1/h5/h20	-2.5 / -2.5 / -1.2	-2.7 / -2.7 / -2.6	-3.9 / -3.0 / -0.9
Best 1-lot passive quote (replay)	1 behind: 45 fills, -1.9/lot, EV -86	1 behind: 30 fills, -2.1/lot, EV -64	at touch: 37 fills, +6.2/lot, EV +230

TELLURIUM (position limit 40)

Archetype: Jump-diffusion; largest and most frequent risk events. **Fair value:** Raw mid; reset immediately after jumps. **Candidate strategy:** Avoid resting size; trade small, wide, and only with jump protection.



Key findings

- 8, 9 and 13 jumps of > 50 per day, mean size 156-178 (max 300), permanent; diffusive tick sigma between jumps is only ~1.6. Direction is unstable (day -2: 7 of 8 down; day -1: 9 of 9 up).
- Most traded product (899-1,600 prints/day). Passive replay loses ~10/lot on day -2 and ~3 on day -1: resting quotes are run over by jumps. A 40-lot position exposed to a 300 jump is ~12,000 of risk.
- Trade-side mix follows the day's direction: 82% of prints at the bid on down day -2, 87% at the ask on up day -1, 50/50 on day 0. This is the only product where flow looks directional; test whether rolling signed flow leads jumps before using it.

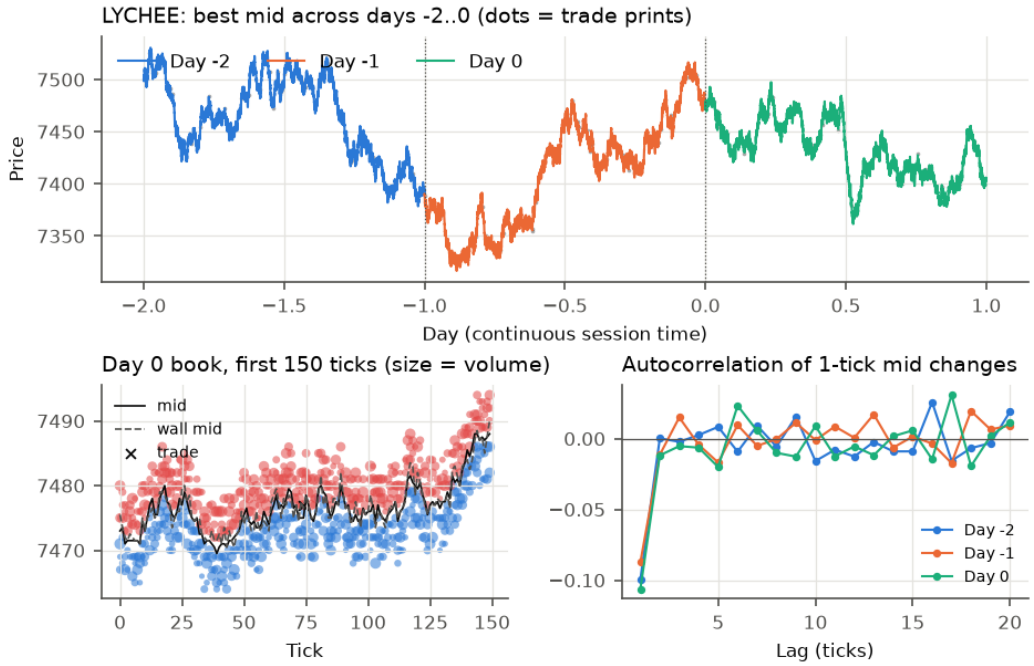
Findings by day

- **Day -2:** fell 1,163.0 (5,001.5 -> 3,838.5, range 3,622.5-5,012.0); spread 9.7, tick sigma 4.93; 8 jumps > 5 sigma. 1176 prints / 4180 lots; aggressor 5-tick markout -2.8. Best passive quote: 1 behind (-5,528 per lot-day).
- **Day -1:** rose 894.0 (3,838.5 -> 4,732.5, range 3,833.0-4,963.0); spread 9.5, tick sigma 5.51; 9 jumps > 5 sigma. 899 prints / 3875 lots; aggressor 5-tick markout -3.4. Best passive quote: 1 behind (-1,094 per lot-day).
- **Day 0:** rose 403.0 (4,731.5 -> 5,134.5, range 4,114.0-5,150.5); spread 9.9, tick sigma 7.06; 13 jumps > 5 sigma. 1600 prints / 6631 lots; aggressor 5-tick markout -3.9. Best passive quote: 1 behind (+320 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	5,001.5 -> 3,838.5 (-1,163.0)	3,838.5 -> 4,732.5 (+894.0)	4,731.5 -> 5,134.5 (+403.0)
Low - high	3,622.5 - 5,012.0	3,833.0 - 4,963.0	4,114.0 - 5,150.5
Mean spread / L1 depth	9.71 / 11.9	9.53 / 12.1	9.92 / 11.8
Tick sigma / zero-move %	4.93 / 15%	5.51 / 15%	7.06 / 15%
ACF(1) mid / wall mid	-0.00 / -0.10	-0.01 / -0.09	-0.01 / -0.06
Jumps > 5 sigma (max move)	8 (260.5)	9 (231.5)	13 (300.5)
Corr momentum10 / EWMA dev -> fwd10	+0.10 / +0.14	-0.00 / -0.01	-0.02 / -0.00
Corr L1 imbalance -> next tick	+0.003	+0.019	-0.006
10-tick fair-value RMSE	mid 16.29; best span 1 -> 16.29	mid 17.29; best span 1 -> 17.29	mid 22.15; best span 1 -> 22.15
Trades / lots (avg size)	1176 / 4180 (3.6)	899 / 3875 (4.3)	1600 / 6631 (4.1)
Prints at ask / bid	18% / 82%	87% / 13%	50% / 50%
Aggressor markout h1/h5/h20	-5.2 / -2.8 / +4.9	-4.6 / -3.4 / -2.4	-4.6 / -3.9 / -4.7
Best 1-lot passive quote (replay)	1 behind: 507 fills, -10.9/lot, EV -5,528	1 behind: 371 fills, -2.9/lot, EV -1,094	1 behind: 681 fills, +0.5/lot, EV +320

LYCHEE (position limit 200)

Archetype: Random walk; underlying for the vouchers. **Fair value:** Raw mid. **Candidate strategy:** Make modestly; primary use is delta hedging the voucher book (limit 200).



Key findings

- Tick sigma ~1.35, spread ~4, slight negative ACF(1) (~ -0.10). Realized vol ~1.6% per day (100-tick returns), below voucher implied vol.
- Deepest L1 book of any product (~25 per side): the best hedge instrument.
- Passive at-touch replay is positive on all days (+22 to +73 per lot-day).

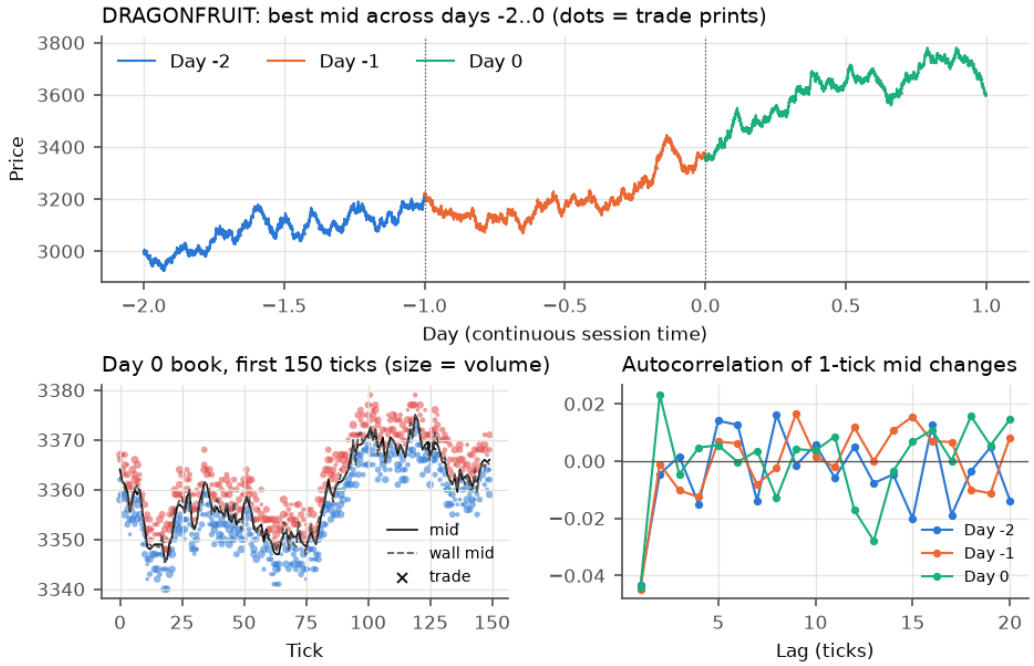
Findings by day

- **Day -2:** fell 109.5 (7,500.0 -> 7,390.5, range 7,378.5-7,530.5); spread 4.0, tick sigma 1.35; no material autocorrelation signal. 53 prints / 119 lots; aggressor 5-tick markout -1.7. Best passive quote: at touch (+72 per lot-day).
- **Day -1:** rose 83.0 (7,390.0 -> 7,473.0, range 7,316.5-7,516.5); spread 4.0, tick sigma 1.35; no material autocorrelation signal. 63 prints / 168 lots; aggressor 5-tick markout -1.9. Best passive quote: at touch (+22 per lot-day).
- **Day 0:** fell 67.5 (7,473.0 -> 7,405.5, range 7,361.5-7,497.5); spread 4.0, tick sigma 1.37; no material autocorrelation signal. 57 prints / 153 lots; aggressor 5-tick markout -2.1. Best passive quote: improve 1 (+60 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	7,500.0 -> 7,390.5 (-109.5)	7,390.0 -> 7,473.0 (+83.0)	7,473.0 -> 7,405.5 (-67.5)
Low - high	7,378.5 - 7,530.5	7,316.5 - 7,516.5	7,361.5 - 7,497.5
Mean spread / L1 depth	4.00 / 25.0	4.00 / 25.0	4.01 / 25.0
Tick sigma / zero-move %	1.35 / 15%	1.35 / 15%	1.37 / 15%
ACF(1) mid / wall mid	-0.10 / -0.34	-0.09 / -0.34	-0.11 / -0.34
Jumps > 5 sigma (max move)	0 (6.0)	0 (5.5)	0 (5.0)
Corr momentum10 / EWMA dev -> fwd10	-0.04 / -0.00	+0.01 / -0.02	-0.05 / -0.00
Corr L1 imbalance -> next tick	+0.004	-0.001	+0.005
10-tick fair-value RMSE	mid 3.90; best span 1 -> 3.90	mid 3.89; best span 1 -> 3.89	mid 3.79; best span 1 -> 3.79
Trades / lots (avg size)	53 / 119 (2.2)	63 / 168 (2.7)	57 / 153 (2.7)
Prints at ask / bid	57% / 43%	62% / 38%	58% / 42%
Aggressor markout h1/h5/h20	-1.8 / -1.7 / -2.7	-2.1 / -1.9 / -1.5	-1.9 / -2.1 / -2.5
Best 1-lot passive quote (replay)	at touch: 35 fills, +2.1/lot, EV +72	at touch: 31 fills, +0.7/lot, EV +22	improve 1: 52 fills, +1.2/lot, EV +60

DRAGONFRUIT (position limit 40)

Archetype: Random walk; basket leg (weight ~2). **Fair value:** Raw mid. **Candidate strategy:** Hedge leg for basket exposure; limited standalone edge.



Key findings

- Trends up every day (+216, +144, +238). ACF(1) ~ -0.04; no imbalance signal.
- Very thin flow (15-23 prints/day) so passive fills are rare and EV is small or negative.
- Return beta in the basket regression ~1.7-1.9 (integer candidate 2).

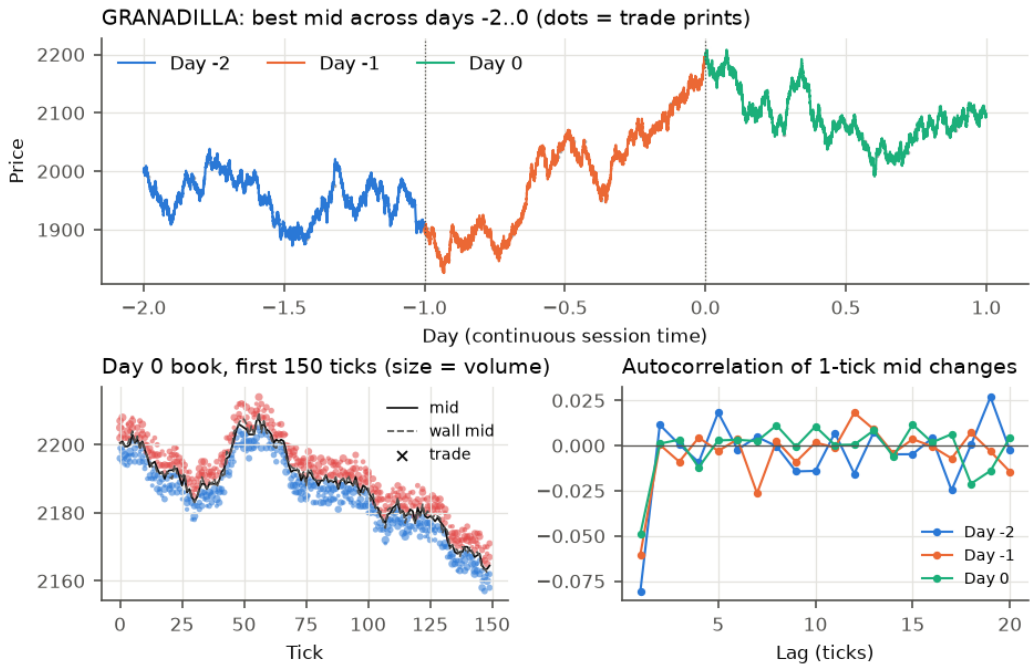
Findings by day

- **Day -2:** rose 216.5 (3,000.0 -> 3,216.5, range 2,924.5-3,216.5); spread 4.0, tick sigma 2.09; no material autocorrelation signal. 23 prints / 48 lots; aggressor 5-tick markout -2.3. Best passive quote: 1 behind (-12 per lot-day).
- **Day -1:** rose 144.5 (3,219.0 -> 3,363.5, range 3,068.0-3,446.5); spread 4.0, tick sigma 2.07; no material autocorrelation signal. 20 prints / 44 lots; aggressor 5-tick markout -2.3. Best passive quote: 1 behind (+14 per lot-day).
- **Day 0:** rose 237.5 (3,364.0 -> 3,601.5, range 3,346.0-3,780.5); spread 4.0, tick sigma 2.07; no material autocorrelation signal. 15 prints / 28 lots; aggressor 5-tick markout -2.7. Best passive quote: 1 behind (+22 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	3,000.0 -> 3,216.5 (+216.5)	3,219.0 -> 3,363.5 (+144.5)	3,364.0 -> 3,601.5 (+237.5)
Low - high	2,924.5 - 3,216.5	3,068.0 - 3,446.5	3,346.0 - 3,780.5
Mean spread / L1 depth	4.01 / 12.5	4.00 / 12.5	4.01 / 12.5
Tick sigma / zero-move %	2.09 / 10%	2.07 / 10%	2.07 / 10%
ACF(1) mid / wall mid	-0.04 / -0.22	-0.04 / -0.23	-0.04 / -0.22
Jumps > 5 sigma (max move)	0 (8.0)	0 (8.0)	0 (8.5)
Corr momentum10 / EWMA dev -> fwd10	-0.01 / +0.00	+0.03 / +0.03	-0.02 / +0.04
Corr L1 imbalance -> next tick	-0.004	+0.010	-0.016
10-tick fair-value RMSE	mid 6.34; best span 1 -> 6.34	mid 6.20; best span 1 -> 6.20	mid 6.43; best span 1 -> 6.43
Trades / lots (avg size)	23 / 48 (2.1)	20 / 44 (2.2)	15 / 28 (1.9)
Prints at ask / bid	39% / 61%	55% / 45%	67% / 33%
Aggressor markout h1/h5/h20	-2.5 / -2.3 / -1.4	-2.2 / -2.3 / -0.4	-2.2 / -2.7 / -3.3
Best 1-lot passive quote (replay)	1 behind: 14 fills, -0.9/lot, EV -12	1 behind: 8 fills, +1.8/lot, EV +14	1 behind: 6 fills, +3.6/lot, EV +22

GRANADILLA (position limit 40)

Archetype: Random walk; basket leg (weight ~3). **Fair value:** Raw mid. **Candidate strategy:** Hedge leg; passive making at touch is mildly positive.



Key findings

- Day moves -99, +300, -105; tick sigma ~1.74; ACF(1) -0.05 to -0.08.
- Basket beta ~2.6-3.2 (integer candidate 3).
- Passive replay at the touch is +32 to +35 per lot-day on all days, on only 10-23 fills.

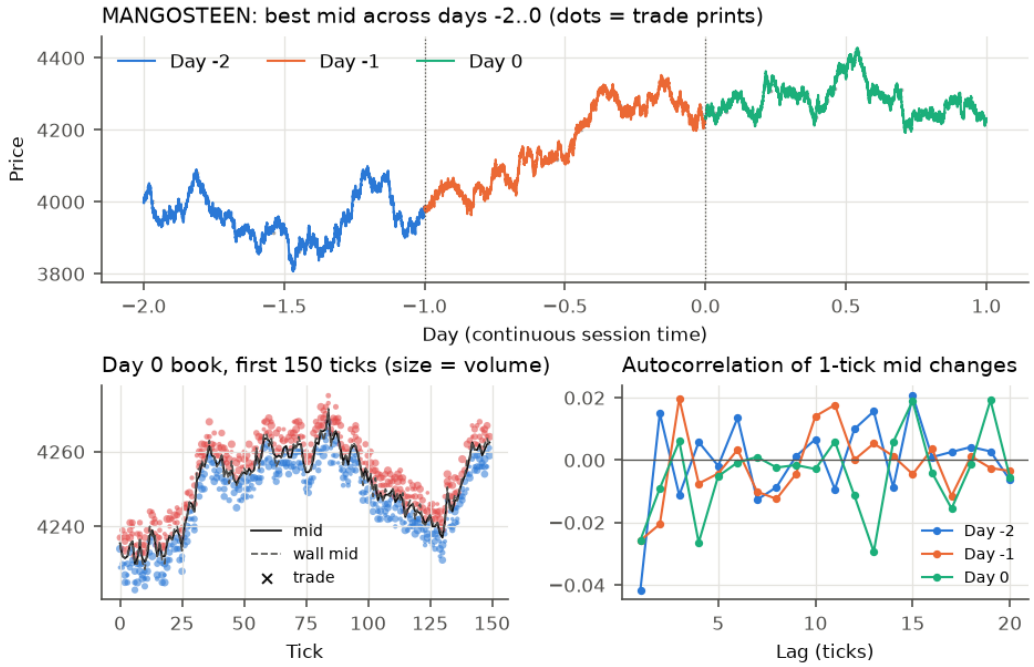
Findings by day

- **Day -2:** fell 99.0 (2,000.0 -> 1,901.0, range 1,872.5-2,038.5); spread 4.0, tick sigma 1.74; no material autocorrelation signal. 23 prints / 44 lots; aggressor 5-tick markout -2.3. Best passive quote: at touch (+34 per lot-day).
- **Day -1:** rose 300.0 (1,901.5 -> 2,201.5, range 1,826.0-2,201.5); spread 4.0, tick sigma 1.74; no material autocorrelation signal. 33 prints / 66 lots; aggressor 5-tick markout -1.4. Best passive quote: 1 behind (+50 per lot-day).
- **Day 0:** fell 105.0 (2,200.5 -> 2,095.5, range 1,991.0-2,208.0); spread 4.0, tick sigma 1.72; no material autocorrelation signal. 19 prints / 41 lots; aggressor 5-tick markout -2.8. Best passive quote: improve 1 (+56 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	2,000.0 -> 1,901.0 (-99.0)	1,901.5 -> 2,201.5 (+300.0)	2,200.5 -> 2,095.5 (-105.0)
Low - high	1,872.5 - 2,038.5	1,826.0 - 2,201.5	1,991.0 - 2,208.0
Mean spread / L1 depth	3.99 / 12.5	4.00 / 12.6	4.00 / 12.5
Tick sigma / zero-move %	1.74 / 11%	1.74 / 11%	1.72 / 12%
ACF(1) mid / wall mid	-0.08 / -0.28	-0.06 / -0.25	-0.05 / -0.27
Jumps > 5 sigma (max move)	0 (6.0)	0 (6.5)	0 (7.0)
Corr momentum10 / EWMA dev -> fwd10	-0.04 / +0.01	-0.01 / -0.01	+0.02 / +0.00
Corr L1 imbalance -> next tick	+0.001	+0.017	-0.003
10-tick fair-value RMSE	mid 5.17; best span 1 -> 5.17	mid 5.13; best span 1 -> 5.13	mid 5.20; best span 1 -> 5.20
Trades / lots (avg size)	23 / 44 (1.9)	33 / 66 (2.0)	19 / 41 (2.2)
Prints at ask / bid	35% / 65%	48% / 52%	37% / 63%
Aggressor markout h1/h5/h20	-1.8 / -2.3 / -2.9	-1.8 / -1.4 / -1.6	-2.1 / -2.8 / -5.2
Best 1-lot passive quote (replay)	at touch: 16 fills, +2.2/lot, EV +34	1 behind: 13 fills, +3.9/lot, EV +50	improve 1: 16 fills, +3.5/lot, EV +56

MANGOSTEEN (position limit 40)

Archetype: Random walk; basket leg (weight ~1). **Fair value:** Raw mid. **Candidate strategy:** Hedge leg; occasional passive making.



Key findings

- Days: -16, +250, -5.5; tick sigma ~2.5; no autocorrelation signal.
- Basket beta ~0.74-0.95 (integer candidate 1).
- Thinnest trade flow in the universe (12-23 prints/day).

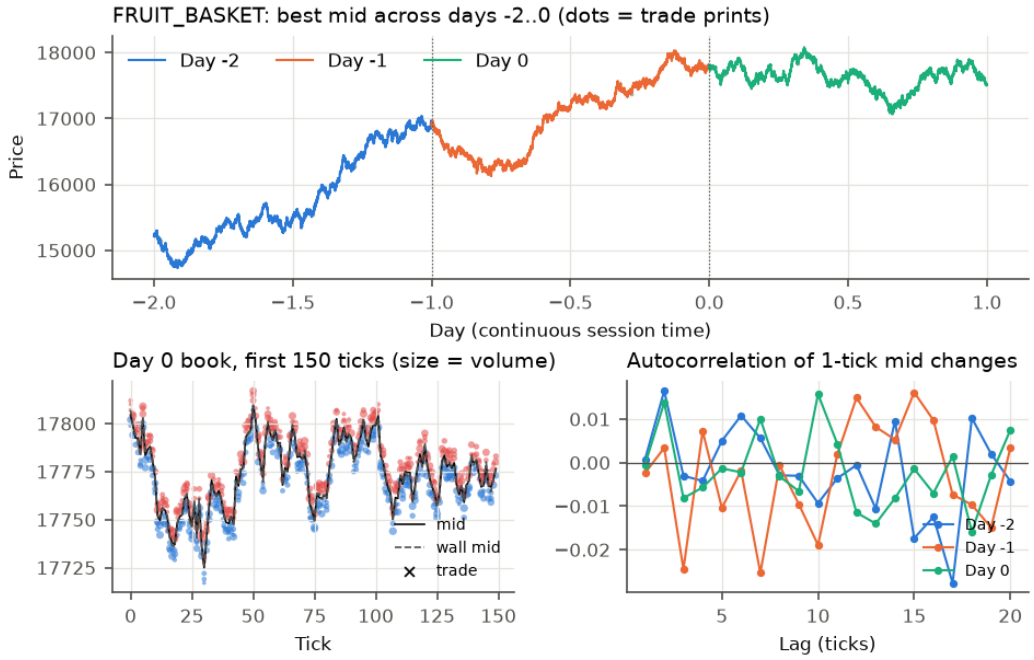
Findings by day

- **Day -2:** fell 16.0 (3,999.5 -> 3,983.5, range 3,807.0-4,099.5); spread 4.0, tick sigma 2.49; no material autocorrelation signal. 12 prints / 21 lots; aggressor 5-tick markout -3.3. Best passive quote: improve 1 (+58 per lot-day).
- **Day -1:** rose 249.5 (3,986.0 -> 4,235.5, range 3,963.0-4,352.5); spread 4.0, tick sigma 2.48; no material autocorrelation signal. 23 prints / 49 lots; aggressor 5-tick markout -2.8. Best passive quote: 1 behind (+34 per lot-day).
- **Day 0:** fell 5.5 (4,235.5 -> 4,230.0, range 4,192.5-4,428.0); spread 4.0, tick sigma 2.47; no material autocorrelation signal. 22 prints / 44 lots; aggressor 5-tick markout -2.1. Best passive quote: at touch (+102 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	3,999.5 -> 3,983.5 (-16.0)	3,986.0 -> 4,235.5 (+249.5)	4,235.5 -> 4,230.0 (-5.5)
Low - high	3,807.0 - 4,099.5	3,963.0 - 4,352.5	4,192.5 - 4,428.0
Mean spread / L1 depth	4.00 / 12.5	4.01 / 12.6	4.01 / 12.5
Tick sigma / zero-move %	2.49 / 8%	2.48 / 8%	2.47 / 8%
ACF(1) mid / wall mid	-0.04 / -0.18	-0.03 / -0.17	-0.03 / -0.17
Jumps > 5 sigma (max move)	0 (9.5)	0 (9.5)	0 (8.5)
Corr momentum10 / EWMA dev -> fwd10	+0.02 / +0.02	+0.01 / -0.04	-0.04 / -0.03
Corr L1 imbalance -> next tick	-0.005	-0.001	+0.002
10-tick fair-value RMSE	mid 7.62; best span 1 -> 7.62	mid 7.57; best span 1 -> 7.57	mid 7.45; best span 1 -> 7.45
Trades / lots (avg size)	12 / 21 (1.8)	23 / 49 (2.1)	22 / 44 (2.0)
Prints at ask / bid	58% / 42%	61% / 39%	50% / 50%
Aggressor markout h1/h5/h20	-2.6 / -3.3 / -5.1	-1.8 / -2.8 / -1.1	-2.5 / -2.1 / -6.6
Best 1-lot passive quote (replay)	improve 1: 11 fills, +5.2/lot, EV +58	1 behind: 12 fills, +2.8/lot, EV +34	at touch: 15 fills, +6.8/lot, EV +102

FRUIT_BASKET (position limit 40)

Archetype: Basket = 2*DRAGONFRUIT + 3*GRANADILLA + 1*MANGOSTEEN + drifting premium. **Fair value:** Synthetic basket plus slowly-updated premium. **Candidate strategy:** Use component prices as a fair-value input; stat-arb on the premium is not supported.



Key findings

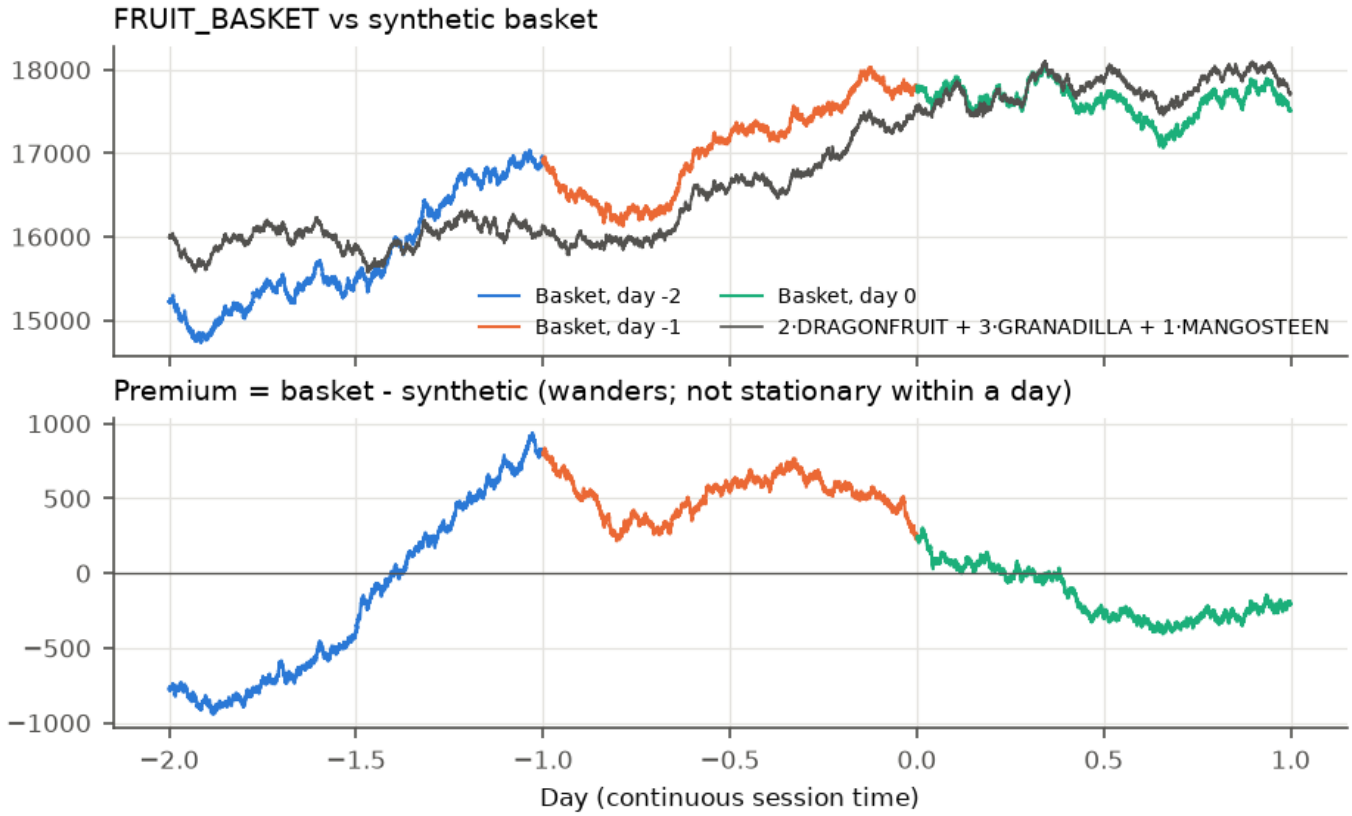
- Basket returns are explained 64-78% by the 2/3/1 synthetic, but the premium is non-stationary: it runs -779 -> +809 on day -2 (ADF p = 0.99) and has a 1,000+ tick half-life on the other days.
- An executable round trip (cross basket and three legs) costs ~30, more than the premium's typical short-horizon movement, so crossing-based arbitrage is not supported by this data.
- Tick sigma ~8 (above the ~7.1 implied by the legs): the basket carries its own idiosyncratic noise.

Findings by day

- **Day -2:** rose 1,707.5 (15,221.0 -> 16,928.5, range 14,731.5-17,040.5); spread 6.0, tick sigma 7.99; no material autocorrelation signal. 15 prints / 32 lots; aggressor 5-tick markout -4.8. Best passive quote: improve 2 (+119 per lot-day).
- **Day -1:** rose 865.5 (16,933.0 -> 17,798.5, range 16,127.5-18,031.0); spread 6.0, tick sigma 7.90; no material autocorrelation signal. 17 prints / 35 lots; aggressor 5-tick markout -5.5. Best passive quote: 1 behind (+24 per lot-day).
- **Day 0:** fell 293.5 (17,804.5 -> 17,511.0, range 17,063.0-18,081.5); spread 6.0, tick sigma 7.96; no material autocorrelation signal. 15 prints / 37 lots; aggressor 5-tick markout -3.4. Best passive quote: improve 2 (-89 per lot-day).

Metric	Day -2	Day -1	Day 0
Open -> close (change)	15,221.0 -> 16,928.5 (+1,707.5)	16,933.0 -> 17,798.5 (+865.5)	17,804.5 -> 17,511.0 (-293.5)
Low - high	14,731.5 - 17,040.5	16,127.5 - 18,031.0	17,063.0 - 18,081.5
Mean spread / L1 depth	6.01 / 12.5	6.00 / 12.5	6.00 / 12.6
Tick sigma / zero-move %	7.99 / 2%	7.90 / 2%	7.96 / 2%
ACF(1) mid / wall mid	+0.00 / -0.01	-0.00 / -0.02	-0.00 / -0.02
Jumps > 5 sigma (max move)	0 (30.0)	0 (31.5)	0 (33.0)
Corr momentum10 / EWMA dev -> fwd10	-0.02 / +0.02	-0.03 / -0.04	-0.02 / -0.00
Corr L1 imbalance -> next tick	+0.013	-0.011	-0.004
10-tick fair-value RMSE	mid 25.76; best span 1 -> 25.76	mid 24.31; best span 1 -> 24.31	mid 25.22; best span 1 -> 25.22
Trades / lots (avg size)	15 / 32 (2.1)	17 / 35 (2.1)	15 / 37 (2.5)
Prints at ask / bid	40% / 60%	59% / 41%	47% / 53%
Aggressor markout h1/h5/h20	-2.2 / -4.8 / -6.8	-4.2 / -5.5 / -6.2	-2.9 / -3.4 / +0.9
Best 1-lot passive quote (replay)	improve 2: 13 fills, +9.2/lot, EV +119	1 behind: 14 fills, +1.8/lot, EV +24	improve 2: 15 fills, -5.9/lot, EV -89

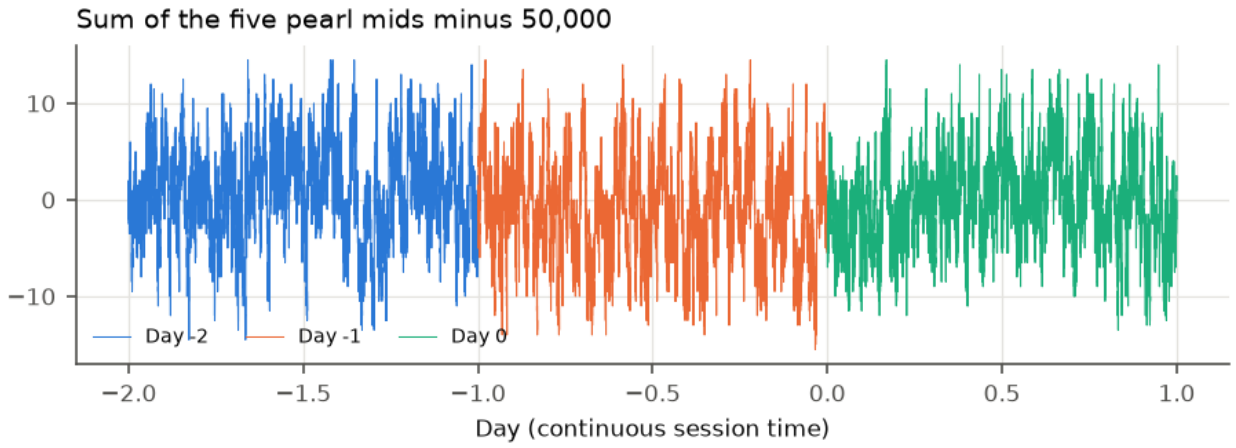
Family: Fruit basket (FRUIT_BASKET vs DRAGONFRUIT, GRANADILLA, MANGOSTEEN)

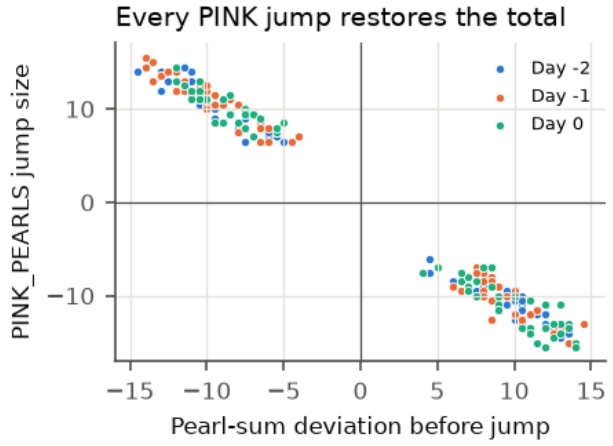


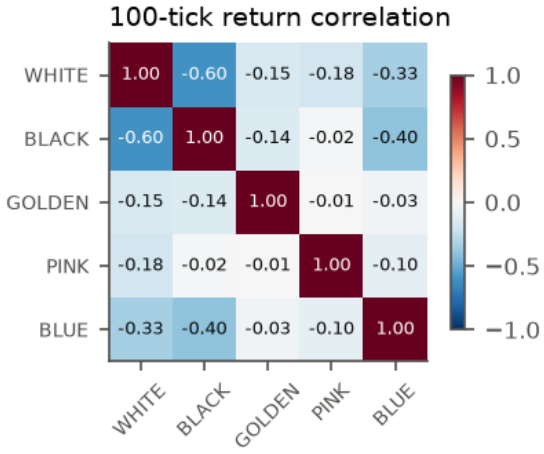
- Regressing 100-tick basket returns on the legs gives betas near (2, 3, 1) every day (R^2 0.64-0.78).
- The premium over 2D + 3G + 1M is NOT mean-reverting within a day (ADF p 0.19-0.99) and moves by hundreds of points; treat it as a random walk.
- Crossing the basket and all three legs costs ~30 per round trip; 40-lot limits and thin leg flow (12-33 prints/day) add legging risk. Do not build a premium stat-arb on three days of evidence.

Day	Beta D	Beta G	Beta M	R^2	Premium start -> end	Premium sd	ADF p	Half-life (ticks)	Round-trip cost
-2	1.91	3.12	0.95	0.73	-778 -> +809	585	0.99	19,319	30.0
-1	1.71	2.64	0.87	0.64	+804 -> +232	136	0.48	1,010	30.0
0	1.90	3.23	0.74	0.78	+240 -> -208	172	0.19	1,652	30.0

Family: Pearls (WHITE, BLACK, GOLDEN, PINK, BLUE)







- The five pearl mids sum to 50,000 +/- 5 (sd 4.6-5.3; ADF p < 0.001; half-life ~10 ticks) on every day. This is the strongest structural relationship in the dataset.
- The sum is not tradable as a basket: the five spreads add up to ~25 against a deviation sd of ~5.
- The constraint is enforced mostly by PINK_PEARLS: its jumps restore the total 100% of the time. GOLDEN is weakly involved (corr -0.05 to -0.13); WHITE/BLACK/BLUE are not.
- Using 50,000 - sum(others) as fair value is WORSE than each pearl's own mid (RMSE ~5 vs 0.5-2.3), so the constraint is a signal for PINK, not a fair-value replacement for the others.

Day	Sum mean	Sum sd	Min - max	ADF p	Half-life	Sum of spreads	PINK jumps	Restoring	P(jump) |dev|>10	P(jump) |dev|<=10
-2	50,000.8	4.85	49,985.5 - 50,014.5	0.000	10.0	25.0	62	100%	8.0%	0.37%
-1	49,998.9	5.29	49,984.5 - 50,014.5	0.000	11.7	25.0	63	100%	5.1%	0.41%
0	49,999.9	4.65	49,986.5 - 50,014.5	0.000	9.1	25.0	63	100%	11.1%	0.39%

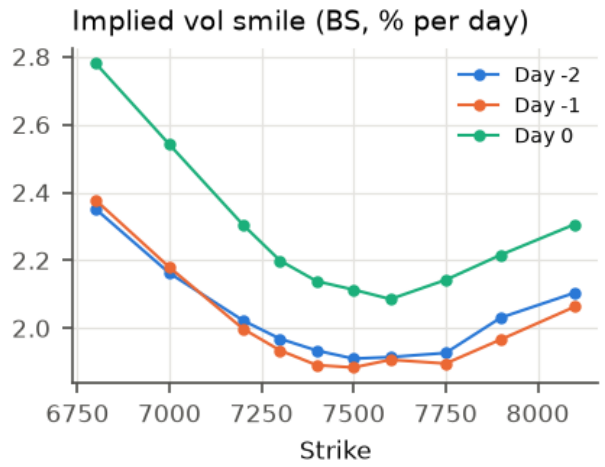
PINK_PEARLS conditional on the pearl-sum deviation

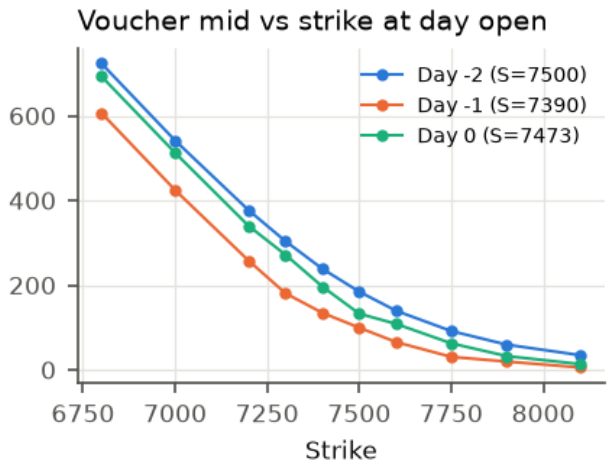
Pearl sum - 50,000	Day -2: mean 20-tick PINK move (n)	Day -1: mean 20-tick PINK move (n)	Day 0: mean 20-tick PINK move (n)
< -10	+6.80 (195)	+4.71 (477)	+5.47 (134)
-10..-5	+2.45 (1079)	+1.87 (2175)	+2.10 (1424)
-5..5	-0.21 (6799)	-0.10 (6012)	-0.01 (6932)
5..10	-2.80 (1722)	-2.48 (1190)	-2.93 (1355)
> 10	-6.62 (185)	-8.92 (126)	-10.19 (135)

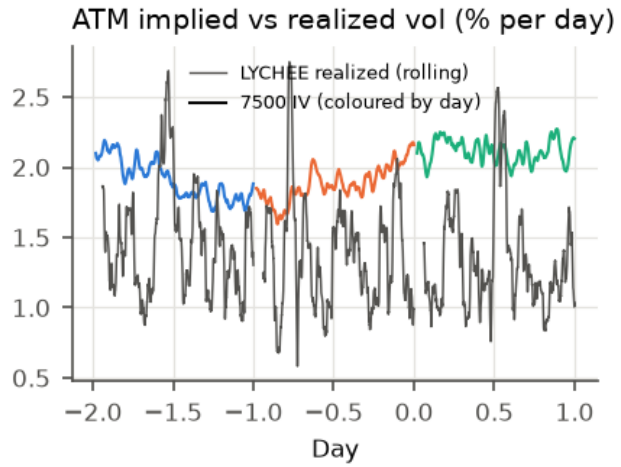
Implied fair value (50,000 - others) vs own mid

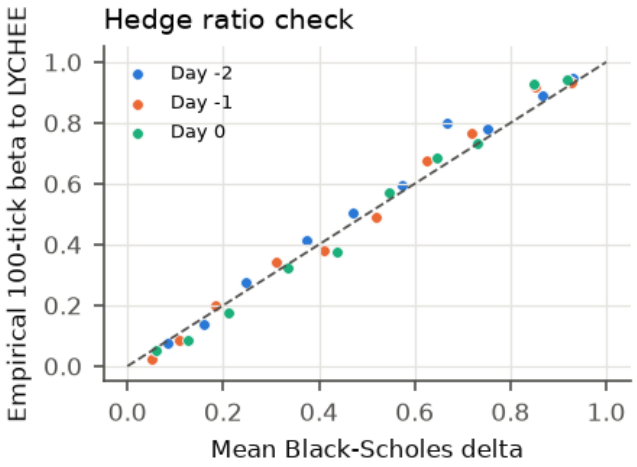
10-tick RMSE	Day -2: own mid / implied	Day -1: own mid / implied	Day 0: own mid / implied
BLACK_PEARLS	7.09 / 8.58	7.07 / 8.98	6.94 / 8.25
BLUE_PEARLS	5.02 / 7.14	5.11 / 7.39	5.03 / 6.87
GOLDEN_PEARLS	0.50 / 4.93	1.46 / 5.43	1.61 / 4.82
PINK_PEARLS	2.78 / 4.53	2.82 / 5.03	2.76 / 4.30
WHITE_PEARLS	7.07 / 8.51	7.00 / 8.82	7.04 / 8.48

Family: LYCHEE vouchers (10 strikes, limit 50 each)









- Ten calls on LYCHEE. Assuming 8/7/6 days to expiry at the open of days -2/-1/0, Black-Scholes IV is ~1.9-2.1% per day at the money, with a clear smile (minimum at 7,500-7,600, rising to ~2.4-2.8% at 6,800).
- IV is ~20-30% above LYCHEE's realized ~1.6%/day on every day: vouchers look rich, which favours net short vega if the expiry convention holds. Day-0 IV is higher at every strike, possibly an expiry-convention artefact; verify how T is decremented.
- No monotonicity, call-spread-width or below-intrinsic violations on executable quotes. Butterfly (convexity) violations are frequent: 3,600-5,500 triplet-ticks/day with mean edge 2-4.5 per unit, concentrated at 7,200-7,600. They come from independent quote noise per strike, not a mispriced curve.
- Empirical 100-tick betas to LYCHEE track BS delta (0.93 deep ITM to 0.05 at 8,100), slightly above it for mid strikes: delta hedging with LYCHEE is viable.
- Highest trade flow of any family (~480-670 prints/day per strike). Passive replay: at-touch quoting is positive for 6,800/7,000 (spread 6) and the wings (7,750-8,100); for 7,200-7,600 (spread 3) only posting behind the touch is reliably positive; improving by 1+ tick is negative.

Static no-arbitrage violations on executable quotes (strike pairs/triplets x ticks)

Day	Monotonicity	Spread > width	Butterfly (convexity)	Ask < intrinsic
-2	0	0	5458	0
-1	0	0	3622	0
0	0	0	4382	0

Voucher findings by strike and day

IV in % per day (BS, T = 8/7/6 days at open, decreasing intraday). Beta = 100-tick regression on LYCHEE. Best quote = best 1-lot passive offset from the replay.

Strike	Day	Open -> close	Spread	Tick sigma	IV %	Beta / BS delta	Trades	Aggr. markout h5	Best passive quote
6800	-2	723.5 -> 607.0 (-116.5)	6.0	1.33	2.35	0.95 / 0.93	477	-2.9	improve 1, EV +584
	-1	606.5 -> 692.5 (+86.0)	6.0	1.33	2.38	0.93 / 0.93	536	-3.0	at touch, EV +858
	0	693.0 -> 622.5 (-70.5)	6.0	1.33	2.78	0.94 / 0.92	604	-2.8	at touch, EV +831
7000	-2	541.5 -> 425.0 (-116.5)	6.0	1.32	2.16	0.89 / 0.87	659	-2.9	improve 1, EV +1,044
	-1	424.5 -> 512.0 (+87.5)	6.0	1.32	2.18	0.92 / 0.85	520	-2.9	at touch, EV +743
	0	512.5 -> 446.5 (-66.0)	6.0	1.32	2.54	0.93 / 0.85	627	-2.8	1 behind, EV +318
7200	-2	376.5 -> 259.5 (-117.0)	3.0	1.20	2.02	0.78 / 0.75	664	-1.3	at touch, EV +168
	-1	257.5 -> 339.5 (+82.0)	3.0	1.16	2.00	0.77 / 0.72	637	-1.4	1 behind, EV +153
	0	339.5 -> 268.5 (-71.0)	3.0	1.17	2.30	0.73 / 0.73	589	-1.3	1 behind, EV +12
7300	-2	303.5 -> 180.5 (-123.0)	3.0	1.17	1.97	0.80 / 0.67	637	-1.1	1 behind, EV +110
	-1	180.5 -> 271.5 (+91.0)	3.0	1.13	1.93	0.68 / 0.63	587	-1.4	at touch, EV +164
	0	271.5 -> 206.5 (-65.0)	3.0	1.12	2.20	0.69 / 0.65	639	-1.4	1 behind, EV +166
7400	-2	238.5 -> 133.5 (-105.0)	3.0	1.14	1.93	0.59 / 0.57	508	-1.4	1 behind, EV +154
	-1	134.5 -> 194.5 (+60.0)	3.0	1.06	1.89	0.49 / 0.52	547	-1.2	1 behind, EV +85
	0	196.5 -> 142.5 (-54.0)	3.0	1.06	2.14	0.57 / 0.54	557	-1.2	1 behind, EV +114
7500	-2	184.5 -> 99.5 (-85.0)	3.0	1.05	1.91	0.51 / 0.47	534	-1.1	1 behind, EV -16
	-1	99.5 -> 131.5 (+32.0)	3.0	0.98	1.88	0.38 / 0.41	480	-1.3	at touch, EV +64
	0	132.5 -> 105.5 (-27.0)	3.0	0.94	2.11	0.38 / 0.44	656	-1.3	at touch, EV +243
7600	-2	139.5 -> 65.5 (-74.0)	3.0	0.98	1.91	0.42 / 0.37	529	-1.1	1 behind, EV +40
	-1	65.5 -> 106.5 (+41.0)	3.0	0.85	1.91	0.34 / 0.31	591	-1.3	at touch, EV +276
	0	108.5 -> 55.5 (-53.0)	3.0	0.84	2.09	0.33 / 0.33	581	-1.4	at touch, EV +220
7750	-2	91.5 -> 31.5 (-60.0)	3.0	0.79	1.93	0.27 / 0.25	524	-1.3	at touch, EV +214
	-1	30.5 -> 62.5 (+32.0)	3.0	0.68	1.90	0.20 / 0.18	546	-1.3	at touch, EV +247
	0	62.5 -> 31.5 (-31.0)	3.0	0.66	2.14	0.18 / 0.21	642	-1.4	at touch, EV +405
7900	-2	59.5 -> 19.5 (-40.0)	3.0	0.65	2.03	0.14 / 0.16	604	-1.4	at touch, EV +406
	-1	19.5 -> 32.5 (+13.0)	3.0	0.54	1.97	0.08 / 0.11	589	-1.4	at touch, EV +537
	0	32.5 -> 20.5 (-12.0)	3.0	0.55	2.22	0.09 / 0.13	547	-1.4	at touch, EV +458
8100	-2	34.5 -> 6.5 (-28.0)	3.0	0.48	2.10	0.08 / 0.08	632	-1.4	at touch, EV +637
	-1	5.5 -> 13.5 (+8.0)	3.0	0.39	2.06	0.02 / 0.05	637	-1.4	at touch, EV +734
	0	13.5 -> 8.5 (-5.0)	3.0	0.40	2.31	0.05 / 0.06	669	-1.4	at touch, EV +764

Open questions and caveats

- M0 (simulator audit) cannot be done from CSVs alone. The passive-quote replay here assumes: an order placed after snapshot t is eligible for trades stamped t+100; one trade fills a 1-lot quote at the submitted price; no queue priority. Validate against the official backtester before relying on EV numbers.
- Voucher time-to-expiry convention (8/7/6 days assumed) and how day 1 is treated.
- Whether the pearl-sum constant (50,000) and PINK's plug behaviour persist out of sample.
- Whether GOLDEN_PEARLS keeps stepping down on day 1, and whether BLUE_PEARLS' uptrend persists.

Three in-sample days are not enough to establish out-of-sample reliability; every finding above is reported per day so its stability can be judged directly.